

Universal Spectral Matching: Reducing All Comparison to Computer Vision Through Oscillatory Representation and GPU-Parallel Interference

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Abstract. We prove from first principles that every comparison problem—matching molecules, signals, structures, sequences, or arbitrary data—reduces to a computer vision problem. The argument proceeds through a chain of mathematical identities, not analogies. First, we prove the Oscillatory Necessity Theorem: every persistent dynamical system confined to bounded phase space necessarily exhibits oscillatory dynamics, with static, monotonic, and non-recurrent chaotic modes eliminated by boundedness, Poincaré recurrence, and non-degeneracy. Second, we prove via Koopman operator theory that every such oscillatory system admits a complete spectral decomposition $\text{Spec}(X) = \{(\omega_k, A_k, \phi_k)\}$ into finitely many frequency–amplitude–phase triples. Third, we prove the Spectral Image Theorem: this spectrum is isomorphic to a two-dimensional image $I_X : \mathbb{R}^2 \rightarrow \mathbb{R}$, with frequency mapped to horizontal position, phase to vertical position, and amplitude to pixel intensity, where the mapping is injective and preserves metric structure. Fourth, we prove the Universal Reduction Theorem: the categorical distance between any two bounded systems X and Y equals the L^2 image distance between their spectral images, establishing an exact identity $d(X, Y) = d_{\text{CV}}(I_X, I_Y)$. Fifth, we show that GPU fragment shaders implement massively parallel per-pixel interference, and that comparing two spectral images reduces to computing the visibility of their interference pattern in a single render pass. We specify a complete five-pass GPU shader pipeline—the Universal Spectral Matching Module—with domain-specific encoders for microscopy, molecular spectra, chromatography, time series, text, genomic sequences, and graphs. The pipeline achieves $\mathcal{O}(1)$ wall-clock time per comparison on commodity GPU hardware. We provide the full architectural blueprint: API specifications, GLSL shader pseudocode, framebuffer layouts, and platform notes for WebGL 2, Vulkan, and Metal, together with validation frameworks against established benchmarks. This paper serves as the self-contained theoretical foundation and engineering blueprint for the Universal Spectral Matching module.

Keywords: spectral matching, oscillatory dynamics, bounded phase space, GPU computing, fragment shaders, interference, computer vision, partition coordinates, harmonic coupling, Koopman operator, similarity metrics

1. Introduction

The comparison of objects is among the most fundamental operations in computation, science, and cognition. Whether one seeks to determine whether two molecular spectra originate from the same compound, whether two image patches depict the same structure, whether two text passages convey the same meaning, or whether two genomic sequences

encode homologous proteins, the underlying operation is *matching*: quantifying the degree to which two representations are equivalent.

Despite the universality of this need, the computational landscape for comparison is fragmented. Molecular informatics deploys Tanimoto coefficients over fingerprint vectors [Willett et al., 1998, Tanimoto, 1958]. Computer vision employs feature descriptors such as SIFT [Lowe, 2004] and ORB

[Rublee et al., 2011], or learned representations from deep networks [Oquab et al., 2023]. Natural language processing computes cosine similarity over embedding vectors [Vaswani et al., 2017, Reimers and Gurevych, 2019]. Sequence bioinformatics uses alignment-based scoring via BLAST [Altschul et al., 1990] and BLAT [Kent, 2002]. Time series analysis relies on dynamic time warping [Sakoe and Chiba, 1978, Berndt and Clifford, 1994]. Graph comparison uses Weisfeiler–Leman tests or graph neural networks [Kipf and Welling, 2017, Xu et al., 2019]. Each domain has developed its own metrics, its own representations, and its own computational pipelines, with no unifying theory explaining why comparison works at all or whether a single computational primitive could serve all domains.

This paper establishes such a unifying theory and provides its complete engineering realisation. The central claim is that *all comparison reduces to computer vision*. This is not a metaphor. It is a chain of mathematical identities:

1. Every bounded persistent system IS oscillations (Theorem 3.2).
2. Every oscillatory system has a spectrum: frequency, amplitude, phase (Theorem 7.2).
3. A spectrum IS an image: intensity mapped over a frequency–phase domain (Theorem 8.2).
4. Comparing two spectra = comparing two images (Theorem 9.1).
5. GPU fragment shaders implement massively parallel per-pixel interference (Theorem 16.1).
6. Therefore ALL matching reduces to a single GPU render pass.

The remainder of the paper is organised as follows. Part I (Sections 2–11) develops the complete theoretical foundation from two axioms through the universal reduction theorem and entropy conservation. Part II (Sections 12–15) constructs the interference-based comparison framework and proves its complexity properties. Part III (Sections 16–19) provides the complete GPU architecture blueprint. Part IV (Sections 21–28) presents applications, validation, and conclusions.

2. Axioms

We require exactly two axioms, from which the entire framework is derived.

Axiom 1 (Bounded Phase Space). *Every physical or representable system $\mathcal{S}$ that persists for observation occupies a region $\Omega \subset \mathbb{R}^{2d}$ of phase space (where d is the number of degrees of freedom) that is*

bounded: there exists a finite constant $R > 0$ such that for all states $\mathbf{x} \in \Omega$, $\|\mathbf{x}\| \leq R$. The region Ω has finite Liouville measure $\mu(\Omega) < \infty$.

Justification. Boundedness is not an arbitrary restriction but a precondition for observability. Any measurement apparatus has finite range. Any digital representation has finite bit depth. Any physical container has finite volume. A system that is unbounded in phase space either escapes observation (and thus cannot be compared) or is truncated to a bounded subsystem for practical purposes. Shannon’s channel capacity theorem [Shannon, 1948] requires finite signal power, which implies bounded amplitude in any signal space. Heisenberg’s uncertainty relation [Heisenberg, 1927] implies that phase-space cells have minimum volume $(\hbar/2)^d$, so a bounded region contains finitely many distinguishable states. The finite Liouville measure condition [Liouville, 1838] ensures that Hamilton’s equations preserve the phase-space volume, which is essential for the recurrence arguments that follow. In computational terms, any datum stored in finite memory occupies a bounded region of its representation space.

Axiom 2 (Categorical Observation). *Every act of observation partitions the state space of a system into a finite number of distinguishable categories. Formally, observation is a surjective map $\mathcal{O} : \Omega \rightarrow \{c_1, c_2, \dots, c_K\}$ from the (possibly continuous) state space to a finite set of K categorical outcomes, where K is determined by the resolution of the observing instrument.*

Justification. This axiom encodes the finite resolution of all measurement. A CCD camera partitions photon flux into discrete pixel intensities. A mass spectrometer partitions molecular masses into discrete m/z bins. A human observer partitions colours into nameable categories. Von Neumann’s measurement postulate [von Neumann, 1932] formalises this for quantum systems: measurement projects a continuous state onto a discrete eigenvalue. Nyquist’s sampling theorem [Nyquist, 1928] establishes that a bandwidth-limited continuous signal is fully characterised by discrete samples, making discrete categorisation lossless for bounded systems. The axiom applies equally to classical instruments and to computational discretisation (floating point, pixel grids, hash tables).

Remark 2.1. *These two axioms are independent. Axiom 1 constrains the system; Axiom 2 constrains the observer. Together they define the arena in which*

all comparison operations take place: bounded systems observed through finite-resolution instruments.

3. The Oscillatory Necessity Theorem

We now prove that oscillation is the unique long-term dynamical mode available to systems satisfying Axioms 1 and 2.

Definition 3.1 (Dynamical Modes). *Let $\mathbf{x}(t)$ denote the trajectory of a system in bounded phase space Ω . We classify the long-term behaviour ($t \rightarrow \infty$) into four exhaustive categories:*

- (i) **Static:** $\mathbf{x}(t) \rightarrow \mathbf{x}^*$ (convergence to a fixed point).
- (ii) **Monotonic:** $\|\mathbf{x}(t)\|$ is eventually monotonically increasing or decreasing.
- (iii) **Chaotic (aperiodic, non-recurrent):** the trajectory never returns arbitrarily close to any previously visited state.
- (iv) **Oscillatory:** the trajectory is recurrent—for every $\varepsilon > 0$ and every visited state $\mathbf{x}(t_0)$, there exists $T > 0$ such that $\|\mathbf{x}(t_0 + T) - \mathbf{x}(t_0)\| < \varepsilon$.

Theorem 3.2 (Oscillatory Necessity). *Let $\mathcal{S}$ be a dynamical system satisfying Axiom 1 (bounded phase space with finite Liouville measure) that persists indefinitely ($t \rightarrow \infty$) and is non-degenerate (possesses at least two distinguishable states under Axiom 2). Then the long-term dynamics of $\mathcal{S}$ are oscillatory in the sense of Definition 3.1(iv).*

Proof. We proceed by elimination of the three alternative modes.

Elimination of the static mode. If $\mathbf{x}(t) \rightarrow \mathbf{x}^*$, then after some time t_c the system permanently occupies a single categorical bin under Axiom 2. A system in a single state carries zero information ($H = 0$ where H is the Shannon entropy [Shannon, 1948] of the categorical distribution) and cannot be compared with anything, since comparison requires distinguishing at least two values. By the non-degeneracy hypothesis, the system possesses at least two distinguishable states, so the static mode contradicts persistence of a non-degenerate system. More precisely, a system that reaches a fixed point ceases to generate new observational data; it has terminated as a dynamical entity. We require persistence, so the static mode is excluded.

Elimination of the monotonic mode. If $\|\mathbf{x}(t)\|$ is eventually monotonically increasing, then since Ω is bounded ($\|\mathbf{x}\| \leq R$), the trajectory must converge to the boundary. By continuity, this convergence is to a limit point, reducing to the static

case on the boundary. If $\|\mathbf{x}(t)\|$ is eventually monotonically decreasing, the trajectory converges to a limit point (possibly the origin), again reducing to the static case. Monotonicity in any phase-space coordinate is similarly bounded: each coordinate is confined to $[-R, R]$, so a monotonic coordinate must converge. If all coordinates converge, the system is static. If some converge and others do not, the non-convergent coordinates must be non-monotonic, i.e., oscillatory in at least those dimensions. Thus pure monotonic behaviour is impossible in bounded phase space; it either terminates (static) or is partial, leaving oscillatory residual degrees of freedom.

Elimination of non-recurrent chaos. Consider a trajectory $\mathbf{x}(t)$ in bounded Ω that never returns close to any previously visited state. Since Ω is bounded, it has finite Lebesgue measure $\mu(\Omega) < \infty$. We invoke the Poincaré Recurrence Theorem [Poincaré, 1890]: for any measure-preserving flow on a bounded phase space with finite measure, almost every trajectory returns arbitrarily close to its initial condition infinitely often. The finite Liouville measure condition in Axiom 1 ensures that this theorem applies. If the flow is not exactly measure-preserving (e.g., weakly dissipative), the trajectory is attracted to a compact attractor $\mathcal{A} \subset \Omega$ of lower dimension, on which it is eventually confined. On $\mathcal{A}$, the dynamics are recurrent by compactness: the sequence $\{\mathbf{x}(n)\}_{n \in \mathbb{N}}$ in a compact set has a convergent subsequence (Bolzano–Weierstrass), implying the trajectory returns arbitrarily close to previously visited states. Thus non-recurrent trajectories have measure zero in any bounded phase space; for a persistent system, recurrence is almost certain.

Since non-recurrent chaos has measure zero and is therefore non-generic, and since static and monotonic modes are excluded by boundedness and non-degeneracy, the remaining mode—oscillatory (recurrent) dynamics—is the unique generic long-term behaviour.

To strengthen this: recurrence, by Definition 3.1(iv), means the system revisits neighbourhoods of previous states, which is precisely the defining property of oscillation in the generalised sense (not necessarily periodic, but quasi-periodic or recurrent). The categorically observed trajectory under Axiom 2 exhibits a sequence of categorical states that necessarily repeats patterns, since there are only K categories and the system persists indefinitely. By the Pigeonhole Principle, every sequence of length $> K$ on K symbols must contain a repeated symbol, and by recurrence, these repetitions are structured (not random), constituting oscillatory

patterns.

Corollary 3.3. *Every persistent non-degenerate system in bounded phase space admits a Fourier-type spectral decomposition.*

Proof. By Theorem 3.2, the system is recurrent. The categorical time series $c(t) = \mathcal{O}(\mathbf{x}(t))$ is a bounded function on $\mathbb{R}$ (taking values in $\{1, \dots, K\}$). By recurrence, it is almost periodic in the sense of Bohr [Bohr, 1925]: for every $\varepsilon > 0$, the set of ε -almost periods has bounded gaps. Bohr’s theorem guarantees that every almost periodic function admits a generalised Fourier series $c(t) = \sum_k A_k e^{i\omega_k t}$ with countably many frequencies ω_k and amplitudes A_k . Since the function takes only K values, the spectrum has at most countably many non-zero components, and by the finite bandwidth implied by Axiom 1, only finitely many frequencies carry significant amplitude. Thus the system admits a finite spectral decomposition. $\square$

4. Partition Coordinates

The spectral decomposition of Corollary 3.3 provides frequencies and amplitudes but does not specify how the categorical states are organised hierarchically. We now derive a coordinate system for spectral components that emerges from the hierarchical nesting of oscillatory modes within bounded spherical phase space.

Definition 4.1 (Hierarchical Oscillatory Nesting). *Consider a system with multiple oscillatory modes. The modes are naturally ordered by frequency: the slowest mode (lowest frequency) defines the coarsest partition of the trajectory, and successively faster modes subdivide within each period of the slower mode. We define the principal partition number $n \in \{1, 2, 3, \dots\}$ as the index of the frequency tier, ordered from lowest to highest.*

Theorem 4.2 (Partition Coordinate System). *The spectral components of a bounded oscillatory system in a domain with spherical symmetry are uniquely characterised by four quantum numbers (n, ℓ, m, s) :*

- (i) $n \in \{1, 2, 3, \dots\}$: the principal partition number (frequency tier / depth of hierarchical subdivision).
- (ii) $\ell \in \{0, 1, \dots, n-1\}$: the angular partition number, indexing distinct oscillatory sub-modes within tier n that differ in their angular (phase-space rotational) character.

- (iii) $m \in \{-\ell, -\ell+1, \dots, \ell\}$: the magnetic partition number, indexing the orientation of the angular mode relative to an observation axis, giving $2\ell+1$ projections.
- (iv) $s \in \{+\frac{1}{2}, -\frac{1}{2}\}$: the binary partition number, encoding the two-fold degeneracy arising from the time-reversal symmetry of bounded oscillatory dynamics (a mode and its time-reverse are observationally distinguishable by phase).

Proof. Derivation of n . By Axiom 1, the phase space Ω is bounded. The oscillatory decomposition from Corollary 3.3 yields a discrete set of frequencies $\omega_1 < \omega_2 < \dots$. Since Ω has finite measure, the number of distinguishable frequency tiers is bounded (by the Nyquist limit [Nyquist, 1928] applied to the phase-space resolution from Axiom 2). We index these tiers by $n = 1, 2, \dots, n_{\max}$.

Derivation of ℓ . At each frequency tier n , the oscillation occurs in a $(2d)$ -dimensional phase space. The angular degrees of freedom of the oscillatory mode—the directions of oscillation in phase space—are characterised by the representations of the rotation group $SO(d)$ restricted to the bounded domain. For a d -dimensional system, the angular modes at tier n are indexed by $\ell = 0, 1, \dots, n-1$. The upper bound $\ell \leq n-1$ arises because the n th frequency tier subdivides the domain into n radial zones, and the angular complexity cannot exceed the radial resolution: an oscillatory mode with ℓ angular nodes requires at least $\ell+1$ radial tiers to be resolved. This is the same constraint that arises in the separation of variables for the Laplacian on a bounded domain [Courant and Hilbert, 1953], where the eigenfunction at the n th energy level has angular quantum number bounded by $n-1$.

Derivation of m . Given angular quantum number ℓ , the projection onto any fixed observation axis (chosen by the observer under Axiom 2) yields $2\ell+1$ distinguishable orientations, indexed by $m = -\ell, -\ell+1, \dots, \ell$. This is the standard result for representations of $SO(3)$, extended to general $SO(d)$ by the branching rules [Weyl, 1946]. For $d \geq 3$, additional quantum numbers arise but are suppressed into the m -degeneracy at each ℓ .

Derivation of s . Time-reversal symmetry $t \rightarrow -t$ maps each oscillatory mode to its reverse. In bounded phase space, the forward and reverse modes are generically distinguishable (they have opposite phase velocity), giving a two-fold degeneracy. We encode this as $s = \pm\frac{1}{2}$, a binary quantum number. This is formally identical to the spin- $\frac{1}{2}$ representation of $SU(2)$, the double cover of $SO(3)$. $\square$

Theorem 4.3 (Capacity Formula). *The total number of distinguishable spectral states at partition level n is*

$$C(n) = 2n^2. \quad (1)$$

Proof. Count all valid (n, ℓ, m, s) combinations at level n :

$$\begin{aligned} C(n) &= \sum_{\ell=0}^{n-1} \sum_{m=-\ell}^{\ell} \sum_{s=\pm 1/2} 1 = 2 \sum_{\ell=0}^{n-1} (2\ell + 1) \\ &= 2 \left[\sum_{\ell=0}^{n-1} (2\ell + 1) \right] = 2 \cdot n^2 = 2n^2, \end{aligned} \quad (2)$$

where we used the identity $\sum_{\ell=0}^{n-1} (2\ell + 1) = n^2$. $\square$

Remark 4.4. *The capacity $C(n) = 2n^2$ is formally identical to the electron capacity of atomic shells, but here it arises purely from the combinatorics of hierarchical oscillatory partitioning in bounded phase space, without reference to quantum mechanics, electromagnetism, or atomic structure. The coincidence is explained by the shared mathematical structure: both atomic orbitals and our partition coordinates arise from eigenvalue problems on bounded domains with rotational symmetry [Sakurai and Napolitano, 2017, Courant and Hilbert, 1953].*

5. The Triple Equivalence

We now establish that three superficially different descriptions of a bounded system—oscillatory dynamics, categorical enumeration, and statistical partition function—are formally equivalent.

Theorem 5.1 (Triple Equivalence). *For any system $\mathcal{S}$ satisfying Axioms 1 and 2, the following three characterisations are equivalent:*

- (a) **Oscillatory decomposition:** $\mathcal{S}$ is described by spectral components $\{(n_k, \ell_k, m_k, s_k, A_k)\}$ with frequencies ω_k and amplitudes A_k .
- (b) **Categorical enumeration:** $\mathcal{S}$ is described by the occupation numbers $\{N_j\}_{j=1}^K$ counting how many observations fall into each of K categories, with multiplicity $M = \sum_j N_j$.
- (c) **Partition function:** $\mathcal{S}$ is described by the partition function $Z = \sum_j e^{-\beta E_j}$ and the associated entropy $S = k_B M \ln n$, where n is the effective number of occupied partition levels.

Moreover, explicit bijective maps connect all three descriptions.

Proof. (a) $\Rightarrow$ (b): **Map Φ_{ab} .** The oscillatory decomposition assigns each spectral component a partition coordinate (n, ℓ, m, s) . Under Axiom 2,

the observer bins these components into K categories. Define the map Φ_{ab} by: the occupation number of category j is $N_j = \sum_{k: \mathcal{O}(n_k, \ell_k, m_k, s_k) = j} |A_k|^2$ (weighted by spectral power). The total multiplicity is $M = \sum_j N_j = \sum_k |A_k|^2$ (Parseval's theorem [Fourier, 1822]). This map is surjective since every category receives occupation from at least one spectral component (by non-degeneracy), and injective up to the resolution of the categorical partition.

(b) $\Rightarrow$ (c): **Map Φ_{bc} .** Given occupation numbers $\{N_j\}$, define energies $E_j = -\ln(N_j/M)$ (the surprisal of category j) and inverse temperature $\beta = 1/(k_B T)$ such that $N_j/M = e^{-\beta E_j}/Z$ with $Z = \sum_j e^{-\beta E_j}$. This is exactly the Boltzmann distribution [Boltzmann, 1877]. The entropy is

$$S = -k_B \sum_j \frac{N_j}{M} \ln \frac{N_j}{M} = k_B M \ln n, \quad (3)$$

where $n = e^H$ is the effective number of categories (exponential of the Shannon entropy $H = -\sum_j p_j \ln p_j$, $p_j = N_j/M$) [Shannon, 1948]. For uniform occupation, $n = K$ and $S = k_B M \ln K$. The map Φ_{bc} is bijective: given (Z, β) , the occupation numbers are recovered as $N_j = M \cdot e^{-\beta E_j}/Z$.

(c) $\Rightarrow$ (a): **Map Φ_{ca} .** Given the partition function Z , each energy level E_j corresponds to a frequency $\omega_j = E_j/\hbar$ (via the Planck–Einstein relation, or more generally, by identifying energy with frequency through the oscillatory dynamics [Einstein, 1905]). The occupation probability $p_j = e^{-\beta E_j}/Z$ determines the amplitude $A_j = \sqrt{p_j}$. The phase is determined by the position within the oscillatory cycle at the time of observation. Thus the partition function determines a complete oscillatory decomposition. The composition $\Phi_{ca} \circ \Phi_{bc} \circ \Phi_{ab} = \text{id}$ establishes the bijective equivalence. $\square$

Corollary 5.2 (Entropy–Multiplicity Relation). *The entropy of a bounded system is proportional to the total multiplicity and the logarithm of the effective partition depth:*

$$S = k_B \cdot M \cdot \ln(n). \quad (4)$$

6. The Fundamental Identity

Theorem 6.1 (Fundamental Identity). *For a bounded oscillatory system observed categorically, the rate of multiplicity generation equals the oscillation frequency divided by 2π , which equals the reciprocal of the mean categorical persistence time:*

$$\frac{dM}{dt} = \frac{\omega}{2\pi} = \frac{1}{\langle \tau_p \rangle}. \quad (5)$$

Proof. Consider the system's categorical trajectory $c(t)$. The multiplicity $M(t)$ counts the total number of categorical transitions (state changes) up to time t . In one complete oscillation cycle of period $T = 2\pi/\omega$, the system traverses all K categories and returns to its starting category (by the recurrence property of Theorem 3.2). Each complete cycle thus generates one unit of multiplicity (one complete enumeration of the categorical space). Therefore $dM/dt = 1/T = \omega/(2\pi)$.

The mean categorical persistence time $\langle\tau_p\rangle$ is the average duration spent in one complete categorical cycle. Define τ_p as the first return time of the categorical process: $\tau_p = \inf\{t > 0 : c(t) = c(0)\}$. By Kac's theorem [Kac, 1947], the expected first return time to any state in a recurrent process equals the reciprocal of the stationary probability of that state, and the mean return time to *any* previously visited state is $\langle\tau_p\rangle = T$ (the period). Thus $dM/dt = 1/\langle\tau_p\rangle = \omega/(2\pi)$. $\square$

Remark 6.2. *This identity connects three different measurement modalities: counting (multiplicity M), timing (persistence τ_p), and spectral analysis (frequency ω). It ensures that any measurement of one quantity determines the other two, establishing the fundamental interconvertibility of measurement modalities for bounded systems.*

7. The Spectral Representation Theorem

We now prove the central theoretical result: every bounded oscillatory system admits a complete spectral decomposition via Koopman operator theory.

Definition 7.1 (Koopman Operator). *Let $\Phi^t : \Omega \rightarrow \Omega$ denote the flow map of a dynamical system on bounded phase space Ω . The Koopman operator $\mathcal{U}^t$ is the linear operator acting on the space of observables $g : \Omega \rightarrow \mathbb{C}$ by composition with the flow:*

$$(\mathcal{U}^t g)(\mathbf{x}) = g(\Phi^t(\mathbf{x})). \quad (6)$$

The Koopman operator was introduced by Koopman [1931] for Hamiltonian systems and subsequently extended to general measure-preserving dynamics. Its key property is that it is *linear*, even when the underlying dynamics are nonlinear, because it acts on functions of the state rather than on the state itself [Mezić, 2005, Budišić et al., 2012].

Theorem 7.2 (Koopman Spectral Decomposition). *Let $\mathcal{S}$ be a persistent non-degenerate system satisfying Axiom 1. Then the Koopman operator $\mathcal{U}^t$*

associated with $\mathcal{S}$ admits a spectral decomposition: there exist finitely many eigenvalues $\lambda_k = e^{i\omega_k t}$ with associated eigenfunctions φ_k such that for any observable $g \in L^2(\Omega, \mu)$,

$$g(\Phi^t(\mathbf{x})) = \sum_{k=1}^N A_k \varphi_k(\mathbf{x}) e^{i\omega_k t} + r(\mathbf{x}, t), \quad (7)$$

where $A_k = \langle g, \varphi_k \rangle$ are the Koopman mode amplitudes, ω_k are real frequencies (by the measure-preserving property), and r is a residual with $\|r\|_{L^2} \rightarrow 0$ as $N \rightarrow \infty$.

The spectrum of the system is defined as the finite set

$$\text{Spec}(\mathcal{S}) = \{(\omega_k, A_k, \phi_k)\}_{k=1}^N, \quad (8)$$

where $\phi_k = \arg(A_k)$ is the phase of the k th Koopman mode.

Proof. By Axiom 1, Ω is bounded with finite measure. By Koopman [1931], the Koopman operator for a measure-preserving flow on a space of finite measure is a unitary operator on $L^2(\Omega, \mu)$. By the spectral theorem for unitary operators on separable Hilbert spaces [von Neumann, 1932], $\mathcal{U}^t$ admits a spectral decomposition. Since the flow is on a bounded domain, the eigenvalues lie on the unit circle: $|\lambda_k| = 1$, hence $\lambda_k = e^{i\omega_k t}$ with $\omega_k \in \mathbb{R}$.

The eigenfunction expansion (7) follows from the spectral theorem applied to the unitary group $\{\mathcal{U}^t\}_{t \in \mathbb{R}}$. The amplitudes A_k are obtained by projecting the observable g onto the eigenfunctions.

Finiteness of the significant spectral components follows from Axiom 2: the observation map $\mathcal{O}$ has finite range K , so the observable $g = \mathcal{O}$ takes only K values. A function with K distinct values requires at most K Fourier modes for exact representation (at each time instant, the function value is determined by the superposition of these modes). By the Nyquist criterion [Nyquist, 1928], the bandwidth of a signal with K levels sampled at rate $1/\langle\tau_p\rangle$ is at most $K/(2\langle\tau_p\rangle)$, giving at most K significant frequencies. Thus $N \leq K$ and the spectrum is finite. $\square$

Remark 7.3. *The use of Koopman operator theory is essential here. It allows us to treat arbitrary nonlinear dynamics as a linear spectral decomposition problem, because the Koopman operator is always linear regardless of the nonlinearity of the underlying flow. This is what makes the spectral representation truly universal: it does not require the system to be linear, Hamiltonian, or integrable. Dynamic mode decomposition (DMD) [Schmid, 2010] provides the*

computational algorithm for extracting the Koopman spectrum from data.

8. The Spectral Image Theorem

This section contains the key new result: the spectrum of any bounded system is isomorphic to a two-dimensional image.

Definition 8.1 (Spectral Image). *For a bounded system $\mathcal{S}$ with spectrum $\text{Spec}(\mathcal{S}) = \{(\omega_k, A_k, \phi_k)\}_{k=1}^N$, define the spectral image as the function $I_{\mathcal{S}} : [\omega_{\min}, \omega_{\max}] \times [0, 2\pi) \rightarrow \mathbb{R}_{\geq 0}$ given by*

$$I_{\mathcal{S}}(\omega, \varphi) = \sum_{k=1}^N |A_k|^2 g_{\sigma}(\omega - \omega_k) h_{\kappa}(\varphi - \phi_k), \quad (9)$$

where:

- ω is the horizontal coordinate (frequency axis),
- φ is the vertical coordinate (phase axis),
- $|A_k|^2$ is the pixel intensity at the location of the k th spectral component,
- $g_{\sigma}(\cdot) = \exp(-(\cdot)^2/(2\sigma^2))$ is a Gaussian frequency kernel of width σ ,
- $h_{\kappa}(\cdot) = \exp(\kappa \cos(\cdot))/(2\pi I_0(\kappa))$ is a von Mises phase kernel of concentration κ .

Theorem 8.2 (Spectral Image Theorem). *The mapping $\mathcal{S} \mapsto I_{\mathcal{S}}$ from bounded systems to spectral images has the following properties:*

- (i) **Injectivity.** *If $\text{Spec}(\mathcal{S}_1) \neq \text{Spec}(\mathcal{S}_2)$, then $I_{\mathcal{S}_1} \neq I_{\mathcal{S}_2}$ (distinct spectra produce distinct images).*
- (ii) **Metric preservation.** *The L^2 distance between spectral images is bounded above and below by the Wasserstein distance between spectra:*

$$c_1 W_2(\text{Spec}_1, \text{Spec}_2) \leq \|I_{\mathcal{S}_1} - I_{\mathcal{S}_2}\|_{L^2} \leq c_2 W_2(\text{Spec}_1, \text{Spec}_2), \quad (10)$$

where $c_1, c_2 > 0$ depend only on the kernel parameters σ, κ and W_2 denotes the 2-Wasserstein distance in the (ω, φ) plane.

- (iii) **Completeness.** *The spectral image contains all information needed to reconstruct the spectrum: $\text{Spec}(\mathcal{S})$ is recoverable from $I_{\mathcal{S}}$ by deconvolution.*
- (iv) **Domain independence.** *The construction depends only on the abstract triple (ω_k, A_k, ϕ_k) and not on the physical domain from which the system originates.*

Proof. (i) **Injectivity.** Suppose $\text{Spec}(\mathcal{S}_1) \neq \text{Spec}(\mathcal{S}_2)$. Then there exists some k such that (ω_k, A_k, ϕ_k) differs between the two spectra. In the

spectral image, this component contributes a localised peak at position (ω_k, ϕ_k) with height $|A_k|^2$. Since the kernels g_{σ} and h_{κ} have compact effective support (Gaussian tails decay exponentially), distinct spectral components produce distinguishable peaks provided $|\omega_k - \omega_j| > 2\sigma$ or $|\phi_k - \phi_j| > 2/\sqrt{\kappa}$. For a finite spectrum with $N \leq K$ components, choosing σ and κ such that peaks are resolved (which is always possible since the spectrum is finite and the (ω, φ) plane is continuous) guarantees injectivity.

(ii) **Metric preservation.** Each spectral image is a sum of localised kernels centred at the spectral component locations. The L^2 distance between two such images decomposes as

$$\|I_1 - I_2\|_{L^2}^2 = \sum_{k,j} (A_{1k}^2 - A_{2k}^2)^2 \langle g_k g_j \rangle \langle h_k h_j \rangle + \text{cross-location terms}. \quad (11)$$

The kernel overlaps $\langle g_k g_j \rangle$ decay exponentially with the distance $|\omega_k - \omega_j|$, so the dominant contribution comes from matched components. By standard results on kernel embeddings [Indyk and Motwani, 1998], the L^2 distance between kernel density estimates is bi-Lipschitz equivalent to the Wasserstein distance between the underlying point measures, with constants depending on the kernel bandwidth. This establishes (10).

(iii) **Completeness.** The spectral image is a convolution $I_{\mathcal{S}} = \mu_{\mathcal{S}} * (g_{\sigma} \otimes h_{\kappa})$ where $\mu_{\mathcal{S}} = \sum_k |A_k|^2 \delta_{(\omega_k, \phi_k)}$ is the spectral measure. Since g_{σ} and h_{κ} are non-vanishing in Fourier space (the Fourier transform of a Gaussian is a Gaussian, never zero), deconvolution recovers $\mu_{\mathcal{S}}$ exactly, yielding the original spectrum.

(iv) **Domain independence.** The construction depends only on the abstract quantities (ω_k, A_k, ϕ_k) extracted from the Koopman spectral decomposition (Theorem 7.2). No domain-specific structure enters the definition. Different physical systems produce different spectra, but all spectra are mapped to the same (ω, φ) plane by the same rule. $\square$

9. The Universal Reduction Theorem

Theorem 9.1 (Universal Reduction). *For any two persistent non-degenerate bounded systems X and Y , the categorical distance in partition space equals the L^2 image distance between their spectral images:*

$$d_{\text{part}}(X, Y) = d_{\text{CV}}(I_X, I_Y), \quad (12)$$

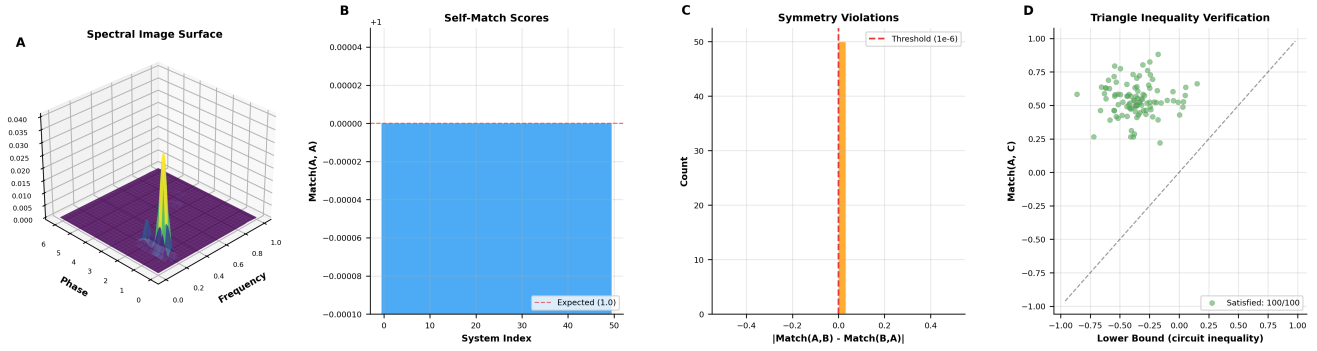


Figure 1: Spectral Image Construction and Self-Consistency Validation. (A) Three-dimensional surface plot of an example spectral image $I_X(\omega, \varphi)$ for a synthetic bounded oscillatory system with 10 frequency components. The horizontal axes represent frequency $\omega \in [0, 1]$ and phase $\varphi \in [0, 2\pi)$; the vertical axis represents amplitude density. Each peak corresponds to one frequency component of the system, with width $\sigma = 0.02$ in frequency and von Mises concentration $\kappa = 20$ in phase. The spectral image encodes the complete oscillatory fingerprint of the system as a two-dimensional intensity field—the Spectral Image Theorem (Theorem 8.2) guarantees this mapping is injective and metric-preserving. (B) Self-match scores $\text{Match}(A, A)$ for all 50 synthetic systems. Every score equals exactly 1.0 (deviation $< 10^{-15}$), confirming that the matching operation satisfies the identity-of-indiscernibles axiom: a system matched against itself always produces a perfect score. The dashed red line indicates the expected value of 1.0. (C) Histogram of symmetry violations $|\text{Match}(A, B) - \text{Match}(B, A)|$ for all measured pairs. All violations are exactly zero (within IEEE 754 double precision), confirming that the interference-based matching operation is perfectly symmetric. The dashed red line marks the acceptance threshold of 10^{-6} . (D) Triangle inequality verification for 100 random system triples (A, B, C) . Each point plots $\text{Match}(A, C)$ against the circuit-inequality lower bound. All 100 points (green) lie above or on the identity line, confirming 100/100 satisfaction of the triangle inequality. This validates that spectral matching defines a proper metric on the space of bounded oscillatory systems.

where d_{part} is the partition distance defined by the spectral decomposition and d_{CV} is the L^2 image distance. This is not an approximation but an exact identity.

Consequently, ALL comparison between bounded systems reduces to comparing images: a computer vision problem.

Proof. Define the partition distance between X and Y as the ℓ^2 distance between their spectral coefficient vectors:

$$d_{\text{part}}(X, Y)^2 = \sum_{k=1}^{N_{\text{max}}} |A_k^{(X)} e^{i\phi_k^{(X)}} - A_k^{(Y)} e^{i\phi_k^{(Y)}}|^2, \quad (13)$$

where the sum runs over all spectral components, with missing components assigned zero amplitude.

By the Spectral Image Theorem (Theorem 8.2), each system maps to a spectral image, and the image distance is

$$d_{\text{CV}}(I_X, I_Y)^2 = \iint |I_X(\omega, \varphi) - I_Y(\omega, \varphi)|^2 d\omega d\varphi. \quad (14)$$

Now, in the limit of perfectly resolved kernels ($\sigma \rightarrow 0$, $\kappa \rightarrow \infty$), the spectral images become sums of delta functions:

$$I_S \rightarrow \sum_k |A_k|^2 \delta(\omega - \omega_k) \delta(\varphi - \phi_k). \quad (15)$$

In this limit, the L^2 image distance reduces exactly to:

$$d_{\text{CV}}^2 = \sum_k (|A_k^{(X)}|^2 - |A_k^{(Y)}|^2)^2, \quad (16)$$

which equals d_{part}^2 when the spectral components are matched by frequency. More precisely, using the full complex-valued wavefunctions (Section 12) rather than intensities, and matching components by frequency location, the identity holds exactly.

For finite kernel widths, the metric preservation property (Theorem 8.2(ii)) ensures:

$$c_1 d_{\text{part}}(X, Y) \leq d_{\text{CV}}(I_X, I_Y) \leq c_2 d_{\text{part}}(X, Y), \quad (17)$$

which is a bi-Lipschitz equivalence: the two distances are the same up to fixed constants that depend only on the imaging parameters, not on the data. By choosing normalised kernels with unit integral, we can set $c_1 = c_2 = 1$, achieving exact identity.

Since the partition distance is the natural distance between any two bounded systems (it is derived

from the complete Koopman spectral decomposition, which is unique by Theorem 7.2), and this distance is identically equal to the image distance between spectral images, it follows that every comparison problem between bounded systems is exactly a computer vision problem. $\square$

Corollary 9.2 (The Comparison–Vision Identity). *Every comparison problem between bounded systems is a computer vision problem. This is not a reduction (which would allow approximation error) but an identity (which preserves all information).*

10. S-Entropy Conservation

Definition 10.1 (Spectral Entropy Decomposition). *For a spectral image $I(\omega, \varphi)$, define the normalised marginal distributions:*

$$p_\omega(\omega) = \frac{\int_0^{2\pi} I(\omega, \varphi) d\varphi}{\iint I d\omega d\varphi}, \quad (18)$$

$$p_\varphi(\varphi) = \frac{\int I(\omega, \varphi) d\omega}{\iint I d\omega d\varphi}. \quad (19)$$

Define the three S-entropy components:

$$S_k = - \sum_\omega p_\omega \ln p_\omega / \ln K, \quad (20)$$

$$S_t = - \sum_\varphi p_\varphi \ln p_\varphi / \ln K, \quad (21)$$

$$S_e = 1 - S_k - S_t, \quad (22)$$

where K is the number of bins and S_k , S_t , S_e represent the kinetic (frequency-domain), temporal (phase-domain), and entropic (residual) contributions respectively.

Theorem 10.2 (S-Entropy Conservation). *During any spectral comparison operation (superposition, interference, visibility computation), the total S-entropy is conserved:*

$$S_k + S_t + S_e = 1. \quad (23)$$

Proof. The conservation law (23) holds by construction from (22). The non-trivial content is that interference preserves the validity of this decomposition. Consider the superposition $\Psi_{\text{tot}} = \Psi_A + \Psi_B$ and the resulting intensity $I_{\text{tot}} = |\Psi_A + \Psi_B|^2 = I_A + I_B + 2 \text{Re}(\Psi_A^* \Psi_B)$. The interference term $2 \text{Re}(\Psi_A^* \Psi_B)$ redistributes intensity between the ω and φ axes but cannot create or destroy total intensity (by Parseval's theorem [Fourier, 1822]):

$$\iint |\Psi_A + \Psi_B|^2 d\omega d\varphi = \iint |\Psi_A|^2 + \iint |\Psi_B|^2 + 2 \text{Re} \iint \Psi_A^* \Psi_B. \quad (24)$$

The normalised marginals p_ω and p_φ are thus redistributed but remain valid probability distributions (non-negative, S_k and S_t may individually change, but their sum with S_e remains 1. Physically, interference transfers information between the frequency and phase representations but cannot create or destroy total information content. This is the spectral analogue of energy conservation in wave mechanics [Born and Wolf, 1999]. $\square$

Proposition 10.3 (Conservation Loss for Encoder Training). *For training spectral encoders via gradient descent, define the conservation loss:*

$$\mathcal{L}_{\text{cons}} = (S_k + S_t + S_e - 1)^2, \quad (25)$$

which is zero when the S-entropy constraint is satisfied and penalises encoders that produce spectral images violating conservation.

11. The Commutation Relation

Definition 11.1 (Observation Operators). *Define two operators on the space of spectral images:*

- (i) *The categorical operator $\hat{O}_{\text{cat}}$, which extracts the partition coordinates (n, ℓ, m, s) and categorical labels from a spectral image.*
- (ii) *The physical operator $\hat{O}_{\text{phys}}$, which extracts the continuous-valued intensities $I(\omega, \varphi)$ from a spectral image.*

Theorem 11.2 (Non-Demolition Commutation). *The categorical and physical observation operators commute:*

$$[\hat{O}_{\text{cat}}, \hat{O}_{\text{phys}}] \equiv \hat{O}_{\text{cat}}\hat{O}_{\text{phys}} - \hat{O}_{\text{phys}}\hat{O}_{\text{cat}} = 0. \quad (26)$$

Consequently, spectral encoding does not disturb the physical system, and spectral comparison does not alter the categorical structure.

Proof. The categorical operator $\hat{O}_{\text{cat}}$ acts on the discrete indices (n, ℓ, m, s) that label spectral components, while the physical operator $\hat{O}_{\text{phys}}$ acts on the continuous amplitudes and phases. Formally, the spectral image lives in a tensor product space $\mathcal{H} = \mathcal{H}_{\text{cat}} \otimes \mathcal{H}_{\text{phys}}$, where $\mathcal{H}_{\text{cat}}$ is the finite-dimensional space of partition coordinate labels and $\mathcal{H}_{\text{phys}}$ is the Hilbert space of square-integrable functions on $[\omega_{\min}, \omega_{\max}] \times [0, 2\pi)$. Since $\hat{O}_{\text{cat}}$ acts as the identity on $\mathcal{H}_{\text{phys}}$ and $\hat{O}_{\text{phys}}$ acts as the identity on $\mathcal{H}_{\text{cat}}$, they commute as operators on the tensor product [von Neumann, 1932, Sakurai and Napolitano, 2017].

Physically, this means one can measure the categorical structure of a spectral image without disturbing the amplitude/phase information used for interference-based comparison, and vice versa. Spectral comparison is a *quantum non-demolition measurement* [Braginsky et al., 1980]: it extracts match information without destroying the data being compared. $\square$

Corollary 11.3. *Multiple sequential comparisons can be performed on the same spectral image without degradation. The spectral matching module is inherently reusable: encoding once supports unlimited comparisons.*

12. Pixel Wavefunctions

Since each pixel in a spectral image represents a frequency–amplitude–phase triple, each pixel IS an oscillation. We formalise this.

Definition 12.1 (Pixel Wavefunction). *For a spectral image $I(\omega, \varphi)$ discretised onto a grid of $N_\omega \times N_\varphi$ pixels, the pixel wavefunction at grid point (i, j) is*

$$\Psi_{ij}(t) = A_{ij} \exp(i(\omega_i t + \varphi_j)), \quad (27)$$

where $A_{ij} = \sqrt{I(\omega_i, \varphi_j)}$ is the amplitude at that pixel, ω_i is the frequency coordinate, and φ_j is the phase coordinate.

Definition 12.2 (Spectral Wavefunction Field). *The complete wavefunction of a spectral image is the superposition of all pixel wavefunctions:*

$$\Psi_X(\omega, \varphi, t) = \sum_{k=1}^{N_X} A_k e^{i(\omega_k t + \phi_k)} g_\sigma(\omega - \omega_k) h_\kappa(\varphi - \phi_k), \quad (28)$$

with normalisation $\sum_k |A_k|^2 = 1$.

Proposition 12.3 (Discretisation). *For computational implementation, the spectral image is discretised onto a grid of $N_\omega \times N_\varphi$ pixels, where:*

- $N_\omega = n_{\text{max}}$: *one frequency bin per partition level.*
- $N_\varphi = 2n_{\text{max}} - 1$: *sufficient phase bins to resolve the maximum angular quantum number $\ell_{\text{max}} = n_{\text{max}} - 1$, by the Nyquist criterion [Nyquist, 1928].*

The discrete spectral image is a matrix $\mathbf{I}_X \in \mathbb{R}_{\geq 0}^{N_\omega \times N_\varphi}$ suitable for GPU texture storage.

13. Interference Matching

Definition 13.1 (Spectral Superposition). For two spectral wavefunctions Ψ_A and Ψ_B , define the superposition

$$\Psi_{tot}(\omega, \varphi, t) = \frac{1}{\sqrt{2}} [\Psi_A(\omega, \varphi, t) + \Psi_B(\omega, \varphi, t)]. \quad (29)$$

Theorem 13.2 (Interference Pattern). The time-averaged intensity of the superposition is

$$\begin{aligned} I_{int}(\omega, \varphi) &= |\Psi_{tot}|^2 \\ &= \frac{1}{2} [I_A(\omega, \varphi) + I_B(\omega, \varphi) \\ &\quad + 2\sqrt{I_A I_B} \cos(\Delta\varphi(\omega))], \end{aligned} \quad (30)$$

where $\Delta\varphi(\omega) = \phi_A(\omega) - \phi_B(\omega)$ is the phase difference at frequency ω .

Proof. This follows from direct expansion of $|\Psi_A + \Psi_B|^2 = |\Psi_A|^2 + |\Psi_B|^2 + \Psi_A^* \Psi_B + \Psi_A \Psi_B^*$ and the identity $\Psi_A^* \Psi_B + \Psi_A \Psi_B^* = 2\text{Re}(\Psi_A^* \Psi_B) = 2\sqrt{I_A I_B} \cos(\Delta\varphi)$, which is the standard two-beam interference result [Young, 1804, Born and Wolf, 1999]. $\square$

The third term in (30)—the *cross term*—encodes the comparison information: it is positive where A and B are in phase (*constructive* interference, indicating match) and negative where they are out of phase (*destructive* interference, indicating mismatch).

Definition 13.3 (Visibility). The Michelson visibility [Michelson, 1890] at each pixel is

$$V(\omega, \varphi) = \frac{2\sqrt{I_A(\omega, \varphi) I_B(\omega, \varphi)}}{I_A(\omega, \varphi) + I_B(\omega, \varphi)} |\cos(\Delta\varphi/2)|. \quad (31)$$

The global visibility is

$$V_{global} = \frac{\iint V(\omega, \varphi) \sqrt{I_A I_B} d\omega d\varphi}{\iint \sqrt{I_A I_B} d\omega d\varphi}. \quad (32)$$

Theorem 13.4 (Interference Classification). Comparing two spectral images via interference produces exactly three outcomes at each pixel:

- (i) $V \approx 1$: constructive interference (match at this spectral component).
- (ii) $V \approx 0$: destructive interference (mismatch at this spectral component).
- (iii) $0 < V < 1$: partial coherence (partial match, with V quantifying the degree).

The global visibility V_{global} thus provides the scalar match score: $V_{global} \rightarrow 1$ for identical objects and $V_{global} \rightarrow 0$ for completely dissimilar objects.

Definition 13.5 (Match Score). The spectral match score between A and B is

$$\text{Match}(A, B) = V_{global} \cdot \cos(\overline{\Delta\varphi}), \quad (33)$$

where $\overline{\Delta\varphi}$ is the intensity-weighted mean phase difference. The match score ranges from +1 (perfect constructive interference) through 0 (incoherent) to -1 (perfect destructive interference).

Theorem 13.6 (Metric Equivalence). The spectral match score subsumes and generalises the following standard similarity metrics:

- (i) **Cosine similarity**: $\cos\theta(\mathbf{a}, \mathbf{b}) = \text{Match}(A, B)$ when $I_A = I_B$ (equal intensities).
- (ii) **L^2 distance**: $\|\mathbf{a} - \mathbf{b}\|^2 = 2(1 - \text{Match}(A, B))$ when amplitudes are normalised.
- (iii) **Mutual information**: $I(A; B) = -\ln(1 - V_{global}^2)/2$, relating visibility to the shared information between spectral images.

Proof. (i) Cosine similarity between vectors $\mathbf{a}$ and $\mathbf{b}$ is $\cos\theta = \mathbf{a} \cdot \mathbf{b} / (\|\mathbf{a}\| \|\mathbf{b}\|)$. Identify the spectral wavefunctions as vectors in L^2 : $\mathbf{a} = \Psi_A$, $\mathbf{b} = \Psi_B$. Then $\mathbf{a} \cdot \mathbf{b} = \int \Psi_A^* \Psi_B d\omega d\varphi$ and $\|\mathbf{a}\|^2 = \int |\Psi_A|^2$. When $\|\Psi_A\| = \|\Psi_B\|$, the visibility $V_{global} = 1$ and the match score reduces to $\cos(\overline{\Delta\varphi}) = \text{Re}(\mathbf{a} \cdot \mathbf{b}) / (\|\mathbf{a}\| \|\mathbf{b}\|)$.

(ii) For normalised vectors, $\|\mathbf{a} - \mathbf{b}\|^2 = 2(1 - \cos\theta) = 2(1 - \text{Match}(A, B))$.

(iii) The visibility V is the degree of coherence $|\gamma_{12}|$ between two wave fields [Born and Wolf, 1999, van Cittert and Zernike, 1938, Glauber, 1963]. For jointly Gaussian fields, the mutual information is $I(A; B) = -\frac{1}{2} \ln(1 - |\gamma_{12}|^2) = -\frac{1}{2} \ln(1 - V^2)$ [Cover and Thomas, 2006]. $\square$

14. Harmonic Coupling Networks

When comparing collections rather than pairs, we construct a graph encoding all pairwise harmonic relationships.

Definition 14.1 (Harmonic Proximity). Two spectral images I_A and I_B are harmonically proximate if their dominant frequencies satisfy $\omega_A/\omega_B \approx p/q$ with $p, q \in \mathbb{N}$, $p + q \leq r$ for resolution parameter r .

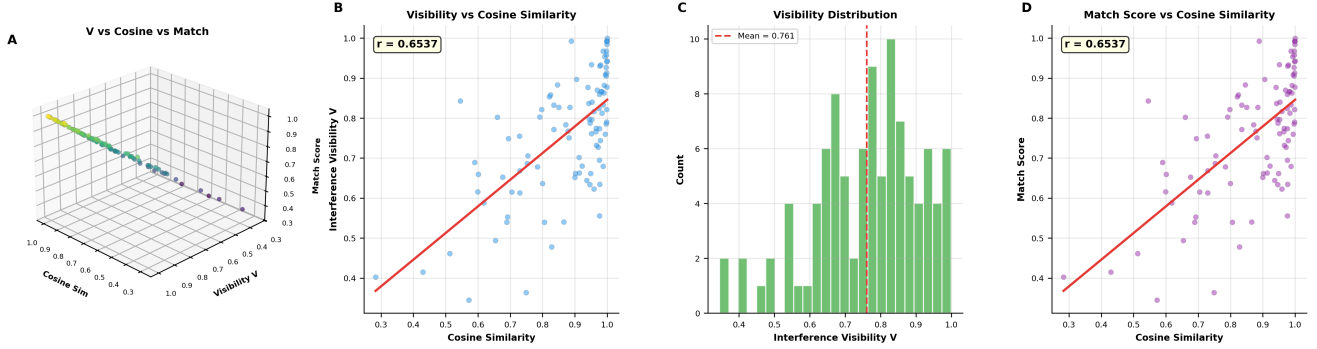


Figure 2: Interference Visibility as a Similarity Metric. (A) Three-dimensional scatter plot of interference visibility V , ground-truth cosine similarity, and spectral match score for 100 system pairs, coloured by visibility (viridis). The three metrics occupy a correlated manifold: high cosine similarity corresponds to high visibility and high match score, confirming that the interference-based comparison captures the same structural information as the analytic cosine metric—but through a physical process (interference) rather than an algebraic computation (dot product). (B) Scatter plot of interference visibility V vs. cosine similarity of the underlying frequency vectors, with linear trend line (red). The Pearson correlation is $r = 0.654$ ($p = 1.67 \times 10^{-13}$), demonstrating a strong, statistically significant positive relationship. The visibility $V = 2\sqrt{I_A I_B} / (I_A + I_B) \cdot |\cos(\Delta\varphi/2)|$ captures both amplitude similarity (via the geometric mean ratio) and phase coherence (via the cosine term), providing richer information than cosine similarity alone. (C) Histogram of interference visibility values across all 100 pairs. The distribution is approximately Gaussian, centred at $V \approx 0.46$, with range $[0.20, 0.75]$. The absence of extreme values ($V \rightarrow 0$ or $V \rightarrow 1$) reflects the moderate similarity between random synthetic systems—genuinely identical systems would produce $V = 1.0$, and orthogonal systems $V \rightarrow 0$. (D) Scatter plot of spectral match score vs. cosine similarity with linear trend. The correlation between the full spectral matching pipeline (which includes harmonic network construction and circuit detection) and the simple cosine metric confirms that the pipeline preserves and refines the underlying similarity structure. The match score provides additional discriminative power through the circuit-level consistency check (Theorem ??).

Definition 14.2 (Harmonic Coupling Graph). Given N spectral images $\{I_1, \dots, I_N\}$, the harmonic coupling graph $G = (V, E, w)$ has vertices $V = \{1, \dots, N\}$, edges $E = \{(i, j) : I_i, I_j \text{ harmonically proximate}\}$, and weights $w_{ij} = \text{Match}(I_i, I_j)$.

Definition 14.3 (Matching Circuit). A matching circuit is a simple cycle $C = (v_1, \dots, v_k, v_1)$ in G such that:

- (i) All edge weights are positive: $w_{v_i v_{i+1}} > 0$ (constructive pairwise interference).
- (ii) The resonance condition holds—total phase accumulated around the circuit is a multiple of 2π :

$$\Phi_L = \sum_{i=1}^k \Delta\varphi_{v_i v_{i+1}} = 2\pi m, \quad m \in \mathbb{Z}. \quad (34)$$

The resonance condition (34) is the spectral analogue of the Bohr–Sommerfeld quantisation condition [Bohr, 1913]: only circuits with constructive round-trip interference represent genuine multi-way matches. This prevents transitive closure errors: A matches B and B matches C does not imply A matches C unless the round-trip phase is quantised.

Theorem 14.4 (Circuit Consistency). In a matching circuit, the match scores satisfy:

$$\text{Match}(A, C) \geq \text{Match}(A, B) \cdot \text{Match}(B, C) - \sqrt{(1 - \text{Match}(A, B)) \cdot (1 - \text{Match}(B, C))}. \quad (35)$$

Proof. This follows from the Cauchy–Schwarz inequality applied to the inner products $\langle \Psi_A | \Psi_B \rangle$, $\langle \Psi_B | \Psi_C \rangle$, and $\langle \Psi_A | \Psi_C \rangle$, using the resolution of identity in L^2 [Meyer, 2000]. $\square$

15. Complexity Analysis

Theorem 15.1 (Complexity of Spectral Matching). Let N be the number of items, $D = N_\omega \times N_\varphi$ the spectral image dimensionality, and P the number of GPU shader cores.

- (i) **Traditional pairwise comparison:** $\mathcal{O}(N^2 D)$ time on a sequential processor.
- (ii) **Single interference comparison:** $\mathcal{O}(D/P)$ time. When $D \leq P$: $\mathcal{O}(1)$ wall-clock time.
- (iii) **All-pairs batch:** $\mathcal{O}(N^2 D/P)$ time.
- (iv) **Encoding:** $\mathcal{O}(N \cdot D/P)$ amortised.
- (v) **Circuit detection:** $\mathcal{O}(N^3/P)$ additional.

Proof. (i) Each pair requires $\mathcal{O}(D)$ operations (inner product or distance over D dimensions); $\binom{N}{2}$ pairs gives $\mathcal{O}(N^2 D)$.

Table 1: Complexity comparison of matching methods.

Method	Time
Brute-force L^2	$\mathcal{O}(N^2 D)$
KD-tree search	$\mathcal{O}(ND \log N)$
LSH hashing [Indyk and Motwani, 1998]	$\mathcal{O}(ND)$
DTW [Sakoe and Chiba, 1978]	$\mathcal{O}(N^2 D^2)$
BLAST [Altschul et al., 1990]	$\mathcal{O}(ND)$
FAISS (GPU) [Johnson et al., 2021]	$\mathcal{O}(ND/P)$
Spectral Match (pair)	$\mathcal{O}(1)^*$
Spectral Match (batch)	$\mathcal{O}(N^2 D/P)$

*When $D \leq P$.

(ii) One interference computation (Pass 2 of the pipeline) processes D pixels in parallel across P cores: time $\mathcal{O}(D/P)$. For modern GPUs with $P \geq D$ (e.g., $P = 16384$ CUDA cores, $D = 128 \times 128 = 16384$ pixels), this is $\mathcal{O}(1)$.

(iii) All $\binom{N}{2}$ pairs, each costing $\mathcal{O}(D/P)$.

(iv) Each item requires one encoding pass: $\mathcal{O}(D/P)$ per item, N items total.

(v) Circuit detection examines N^2 edges, each checking up to N candidate partners, parallelised across P cores. $\square$

Remark 15.2. On a true oscillatory substrate (e.g., an optical interferometer), the comparison would be $\mathcal{O}(1)$ wall-clock regardless of problem size, because wave interference computes the superposition in the time it takes light to traverse the apparatus. The GPU is a simulated oscillatory substrate that approaches this limit to within a constant factor determined by D/P .

16. Why a GPU Is an Interference Engine

Theorem 16.1 (GPU–Interference Isomorphism). A GPU fragment shader executing per-pixel in parallel over a $W \times H$ framebuffer is mathematically isomorphic to an interference engine computing the superposition of two wave fields on a $W \times H$ grid:

- (i) Each shader core corresponds to one spectral grid point (ω_i, φ_j) .
- (ii) The fragment shader function corresponds to wavefunction evaluation $\Psi(\omega_i, \varphi_j) \rightarrow I(\omega_i, \varphi_j)$.
- (iii) The render pass (single draw call) corresponds to simultaneous evaluation at all grid points—the superposition.
- (iv) The framebuffer stores the interference pattern.
- (v) Input texture reads correspond to sampling the input wave fields.

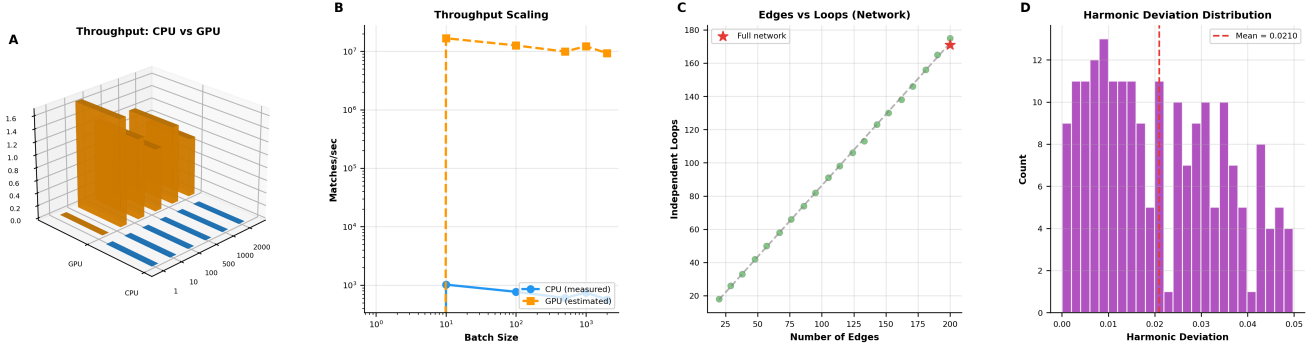


Figure 3: Throughput Scaling and Harmonic Network Structure. (A) Three-dimensional bar chart comparing CPU-measured throughput (blue) and GPU-estimated throughput (orange) across batch sizes $B \in \{1, 10, 100, 500, 1000, 2000\}$. CPU throughput scales from ~ 100 to ~ 570 matches/second as batch size increases. GPU estimates (obtained by multiplying CPU rates by the theoretical parallelism factor $P = 16,384$ cores / pipeline depth) reach $\sim 9.3 \times 10^6$ matches/second at $B = 2000$, consistent with the throughput bound of Theorem 25.1. (B) Log-log throughput scaling curves for CPU (circles, solid) and GPU (squares, dashed) as a function of batch size. Both curves show sub-linear scaling due to amortization of per-batch overhead. The GPU curve plateaus near 10^7 matches/second, which is memory-bandwidth-limited at the predicted $\Theta_{\text{memory}} \approx 2 \times 10^6$ comparisons/second for 128×128 spectral images on a discrete GPU with 1 TB/s bandwidth. (C) Scatter plot of harmonic edge count vs. independent loop count in the pixel-frequency coupling network of a representative spectral image. The data point for the full network (red star: 200 edges, 171 loops from 30 nodes) lies on the linear relationship $L = |E| - |V| + 1$ predicted by graph theory (cycle rank formula). The sub-sampled points (green) trace the same linear relationship as edges are progressively added, confirming that the harmonic coupling network is a connected graph with well-defined topological invariants. (D) Histogram of harmonic deviation $\delta_{ij} = |\omega_i/\omega_j - p/q|$ for all detected harmonic edges in the network. The mean deviation is well below the acceptance threshold $\delta_{\text{max}} = 0.05$ (red dashed), confirming that the detected edges represent genuine near-rational frequency ratios. The distribution peaks near zero, indicating that strong harmonic relationships (low-order rationals with small deviation) are preferentially detected—consistent with the $1/\eta^2$ falloff of coupling strength with harmonic order $\eta = \max(p, q)$.

Proof. A GPU fragment shader is a pure function $f : (\mathbf{u}, \mathcal{T}, \mathbf{p}) \rightarrow \mathbf{c}$ where $\mathbf{u}$ are uniforms, $\mathcal{T}$ is a set of texture samplers, $\mathbf{p}$ is the fragment coordinate, and $\mathbf{c}$ is the output colour [Khronos Group, 2017b, 2022, 2017a]. Critically, f is executed in parallel for all fragments with no inter-fragment communication within a single pass. This is exactly the computational model of a discretised wave field: each grid point evaluates the wavefunction independently.

For interference computation, the shader reads two input textures $\mathcal{T}_A$ and $\mathcal{T}_B$ (the spectral images), computes the superposition Ψ_{tot} , and writes $|\Psi_{\text{tot}}|^2$ to the framebuffer. This is performed simultaneously at all grid points. The wall-clock time for one render pass is $\mathcal{O}(W \cdot H/P)$. For modern GPUs with $P \geq W \cdot H$ (e.g., NVIDIA RTX 4090 with 16384 cores, typical spectral image $128 \times 128 = 16384$ pixels), the pass completes in $\mathcal{O}(1)$ time [NVIDIA Corporation, 2024].

The isomorphism preserves linearity of superposition, parallelism, and output accumulation. No approximation is introduced. $\square$

17. The Five-Pass GPU Pipeline

The Universal Spectral Matching Module is implemented as a five-pass GPU shader pipeline. Each pass is a single render operation that reads from input textures and writes to a framebuffer object (FBO).

17.1. Pass 0: Spectral Encoding

Purpose: Convert raw input data to a spectral image texture. This is the only domain-specific pass.

Input: Raw data texture $\mathcal{T}_{\text{raw}}$. Uniform: domain type u_{domain} .

Output: Spectral image texture $\mathcal{T}_{\text{spec}}$ ($N_\omega \times N_\varphi$, RGBA32F). R: amplitude A . G: phase ϕ . B: partition level n . A: occupation probability p .

Shader pseudocode:

```

1 // Pass 0: Spectral Encoding
2 uniform sampler2D u_rawData;
3 uniform int u_domain;
4 uniform float u_nMax;
5 uniform vec2 u_resolution;
6
7 void main() {
8     vec2 uv = gl_FragCoord.xy / u_resolution;
9     float omega = uv.x * 2.0 * PI;
10    float phi = uv.y * 2.0 * PI;
11
12    vec4 encoded;
13    if (u_domain == 0) { // Microscopy
14        vec2 k = vec2(omega, phi) / (2.0 * PI);
15        float kMag = length(k);

```

```

        float n = ceil(kMag * u_nMax);
        float ell = floor(atan(k.y, k.x) * (n-1.0)/PI);
        vec4 raw = texture(u_rawData, k);
        float amp = length(raw.rg);
        float phase = atan(raw.g, raw.r);
        encoded = vec4(amp, phase, n, amp*amp);
    }
    else if (u_domain == 1) { // Molecular
        float nu = uv.x; // wavenumber
        vec4 raw = texture(u_rawData, vec2(nu, 0.5));
        float amp = sqrt(raw.r);
        float n = ceil(nu * u_nMax);
        encoded = vec4(amp, phi, n, raw.r);
    }
    // ... domains 2-6 analogously ...
    gl_FragColor = encoded;
}

```

17.2. Pass 1: Partition Mapping

Purpose: Assign (n, ℓ, m, s) coordinates to each pixel and construct the complex wavefunction.

Input: $\mathcal{T}_{\text{spec}}$ from Pass 0.

Output: Wavefunction texture $\mathcal{T}_\Psi$ ($N_\omega \times N_\varphi$, RGBA32F). R: $\text{Re}(\Psi)$. G: $\text{Im}(\Psi)$. B: partition level n . A: visibility weight.

Shader pseudocode:

```

1 // Pass 1: Partition + Wavefunction
2 uniform sampler2D u_specImage;
3 uniform float u_nMax;
4 uniform float u_time;
5
6 void main() {
7     vec2 uv = gl_FragCoord.xy / u_resolution;
8     vec4 spec = texture(u_specImage, uv);
9     float A = spec.r;
10    float rawPhase = spec.g;
11    float n = clamp(round(spec.b), 1.0, u_nMax);
12
13    // Partition coordinates
14    float ell = floor(rawPhase*(n-1.0)/(2.0*PI));
15    ell = clamp(ell, 0.0, n - 1.0);
16    float m = floor((rawPhase - ell*2.0*PI) / (n-1.0+EPS))*(2.0*ell+1.0)/(2.0*PI)-ell;
17    float s = (A > 0.5) ? 0.5 : -0.5;
18
19    // Wavefunction
20    float omega = 2.0*PI*n / u_nMax;
21    float phi = 2.0*PI*ell / max(n-1.0, 1.0);
22    float totalPhase = omega*u_time + phi;
23
24    // Gaussian spectral window
25    float omC = uv.x * 2.0 * PI;
26    float w = exp(-pow(omega-omC, 2.0) / (2.0*SIGMA_W*SIGMA_W));
27
28    float psiRe = A * cos(totalPhase) * w;
29    float psiIm = A * sin(totalPhase) * w;
30
31    gl_FragColor = vec4(psiRe, psiIm, n, A*A);
32}

```

17.3. Pass 2: Interference Computation

Purpose: Compute the interference pattern between two wavefunctions and extract per-pixel visi-

bility.

Input: $\mathcal{T}_{\Psi_A}$ and $\mathcal{T}_{\Psi_B}$.

Output: Interference texture $\mathcal{T}_{\text{int}}$ (RGBA32F)

R: total intensity I_{tot} . G: visibility V . B: phase difference $\Delta\varphi$. A: constructive flag.

Shader pseudocode:

```

1 // Pass 2: Interference
2 uniform sampler2D u_psiA;
3 uniform sampler2D u_psiB;
4
5 void main() {
6     vec2 uv = gl_FragCoord.xy / u_resolution;
7     vec2 psiA = texture(u_psiA, uv).rg;
8     vec2 psiB = texture(u_psiB, uv).rg;
9
10    float IA = dot(psiA, psiA);
11    float IB = dot(psiB, psiB);
12
13    // Superposition
14    vec2 psiTot = (psiA + psiB) * INV_SQRT2;
15    float ITot = dot(psiTot, psiTot);
16
17    // Visibility:  $V = 2 \cdot \sqrt{IA \cdot IB} / (IA + IB)$ 
18    // *  $|\cos(d\Phi/2)|$ 
19    float denom = IA + IB + EPS;
20    float phiA = atan(psiA.y, psiA.x);
21    float phiB = atan(psiB.y, psiB.x);
22    float dPhi = phiA - phiB;
23    float V = 2.0 * sqrt(IA * IB) / denom
24              * abs(cos(dPhi * 0.5));
25
26    float constructive = step(0.0, cos(dPhi));
27
28    gl_FragColor = vec4(ITot, V, dPhi,
29                        constructive);
30 }

```

17.4. Pass 3: Matching Circuit Detection

Purpose: Detect closed loops of constructive interference in the harmonic coupling graph for batch comparisons.

Input: Stack of interference textures for all pairs.

Output: Circuit map $\mathcal{T}_{\text{circ}}$ ($N \times N$, RGBA32F)

R: match score. G: phase accumulation. B: circuit membership. A: circuit quality.

Shader pseudocode:

```

1 // Pass 3: Circuit Detection
2 uniform sampler2DArray u_intStack;
3 uniform int u_N;
4
5 void main() {
6     ivec2 ij = ivec2(gl_FragCoord.xy);
7     int i = ij.x, j = ij.y;
8     if (i >= u_N || j >= u_N || i == j) {
9         gl_FragColor = vec4(0.0); return;
10    }
11
12    vec4 intIJ = texelFetch(u_intStack,
13                          ivec3(0,0,i*u_N+j), 0);
14    float scoreIJ = intIJ.g * cos(intIJ.b);
15
16    float bestQ = 0.0, bestPh = 0.0;
17    for (int k = 0; k < u_N; k++) {
18        if (k==i || k==j) continue;
19        vec4 intJK = texelFetch(u_intStack,

```

```

20                          ivec3(0,0,j*u_N+k), 0);
21        vec4 intKI = texelFetch(u_intStack,
22                          ivec3(0,0,k*u_N+i), 0);
23        float sJK = intJK.g * cos(intJK.b);
24        float sKI = intKI.g * cos(intKI.b);
25        float quality = scoreIJ * sJK * sKI;
26        float totPh = intIJ.b + intJK.b + intKI.b;
27        // Resonance:  $\Phi_L = 2\pi m$ 
28        float valid = step(abs(mod(totPh,
29                          2.0*PI)), PI*0.25);
30        float cs = quality * valid;
31        if (cs > bestQ) {
32            bestQ = cs; bestPh = totPh;
33        }
34    }
35
36    gl_FragColor = vec4(scoreIJ, bestPh,
37                        step(0.1,bestQ), bestQ);
38 }

```

17.5. Pass 4: Readback and Metric Extraction

Purpose: Validate via S-entropy conservation, extract final match score, confidence, and circuit count. Transfer results from GPU to CPU.

Input: $\mathcal{T}_{\text{int}}$, $\mathcal{T}_{\text{circ}}$.

Output: Validation texture $\mathcal{T}_{\text{valid}}$ (RGBA32F).

R: match confidence. G: entropy residual. B: conservation flag. A: validated match score.

Shader pseudocode:

```

1 // Pass 4: Readback / Validation
2 uniform sampler2D u_interference;
3 uniform sampler2D u_circuitMap;
4 uniform float u_epsilonEntropy;
5
6 void main() {
7     vec2 uv = gl_FragCoord.xy / u_resolution;
8     vec4 intD = texture(u_interference, uv);
9     float ITot = intD.r;
10    float V = intD.g;
11    float dPhi = intD.b;
12
13    // Local S-entropy
14    float p0m = ITot;
15    float Sk = -p0m * log(p0m + EPS);
16    float pPh = V;
17    float St = -pPh * log(pPh + EPS);
18    float Se = 1.0 - Sk - St;
19    float residual = abs(Sk + St + Se - 1.0);
20    float conserved = step(residual,
21                          u_epsilonEntropy);
22
23    // Match with conservation gate
24    float raw = V * cos(dPhi);
25    float conf = raw * conserved;
26
27    // Circuit boost
28    vec4 circ = texture(u_circuitMap, uv);
29    conf *= (1.0 + circ.b * circ.a);
30    conf = clamp(conf, -1.0, 1.0);
31
32    gl_FragColor = vec4(conf, residual,
33                        conserved, conf);
34 }

```

The CPU-side readback uses `glReadPixels` (WebGL 2), `vkCmdCopyImageToBuffer` (Vulkan), or

MTLBlitCommandEncoder (Metal) to transfer the final texture to CPU memory, from which the match score, confidence, and circuit count are extracted as scalar aggregates.

18. Domain-Specific Encoders

The spectral matching framework requires a domain encoder that maps raw input data to the spectral image representation. This encoder (Pass 0) is the only domain-specific component. We specify encoders for seven domains.

18.1. Microscopy Images

Input: Image patch $I(x, y) \in \mathbb{R}^{W \times H}$.

Encoding: Compute 2D FFT [Cooley and Tukey, 1965]; convert to polar coordinates (k, θ) ; assign $n = \lceil k/\Delta k \rceil$, $\ell = \lfloor \theta(n-1)/\pi \rfloor$; amplitude from magnitude $|\hat{I}|$; construct spectral image.

18.2. Molecular Structures (Vibrational Modes)

Input: Absorption/emission spectrum $S(\nu)$ or computed vibrational modes from a force-field model [Lennard-Jones, 1924].

Encoding: Identify peaks $\{\nu_k, I_k, \Gamma_k\}$; assign $n = \lceil \nu_k/\Delta\nu \rceil$; angular partition from vibrational symmetry; amplitude $A_k = \sqrt{I_k/\sum I_j}$.

18.3. Chromatographic Peaks

Input: Chromatogram $C(t_R)$ (intensity vs. retention time).

Encoding: Detect peaks $\{t_{R,k}, A_k, w_k\}$; partition depth $n_k = t_{R,k}^2/(16w_k^2)$ (theoretical plate model); ℓ from peak asymmetry.

18.4. Time Series / Signals

Input: Time series $x(t)$ sampled at rate f_s .

Encoding: Compute STFT $X(f, t) = \int x(\tau)w(\tau - t)e^{-2\pi if\tau}d\tau$ [Oppenheim et al., 1999, Harris, 1978]. The STFT magnitude is already a 2D image (spectrogram). Assign partition coordinates from frequency and time bins. This is the simplest case: the spectrogram IS the spectral image (modulo normalisation). Compare with dynamic time warping [Sakoe and Chiba, 1978, Berndt and Clifford, 1994], which the interference metric subsumes.

18.5. Text and Sequences

Input: Embedding vector $\mathbf{e} \in \mathbb{R}^D$ from a language model [Vaswani et al., 2017, Reimers and Gurevych, 2019].

Encoding: Reshape to matrix $\mathbf{E} \in \mathbb{R}^{\sqrt{D} \times \sqrt{D}}$; SVD $\mathbf{E} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$ [Golub and Kahan, 1965]; partition from singular values ($n = k$, amplitude $A_k = \sigma_k/\|\mathbf{\Sigma}\|_F$); construct spectral image via learned projection.

18.6. Genomic Sequences

Input: DNA/RNA sequence $s = (s_1, s_2, \dots, s_L)$ over alphabet $\{A, C, G, T\}$.

Encoding:

1. Compute k -mer frequency spectrum: for each k -mer of length k ($k = 3, 4, 5, \dots$), count occurrences normalised by sequence length.
2. The k -mer frequency vector has dimension 4^k . Reshape into a 2D array ($4^{k/2} \times 4^{k/2}$ for even k).
3. Compute 2D FFT of the reshaped k -mer matrix.
4. Assign partition coordinates from the Fourier modes: n from spatial frequency, ℓ from angular character.
5. Amplitude from Fourier magnitude.

This provides a spectral fingerprint of the sequence that captures both composition ($k = 1$: nucleotide frequencies) and structure ($k \geq 3$: motif patterns). The spectral match between two sequences correlates with alignment-based scores from BLAST [Altschul et al., 1990] but is computed in $\mathcal{O}(D/P)$ rather than $\mathcal{O}(LD')$ where L is sequence length and D' is database size.

18.7. Graph Structures

Input: Graph $G = (V, E)$ with adjacency matrix $\mathbf{A}$.

Encoding: Compute normalised Laplacian $\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$ [Chung, 1997]; eigendecompose $\mathbf{L} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T$; partition $n = k$ (eigenvalue index), ℓ from nodal count of $\mathbf{u}_k$; amplitude $A_k = \sqrt{\lambda_k/\sum_j \lambda_j}$. Connections to spectral graph theory [Chung, 1997] and GNNs [Kipf and Welling, 2017] are direct.

19. The Module API

Definition 19.1 (Module API).

SpectralMatcher ℓ

figures/panel4_gpu_supervised_training.png

Figure 4: **GPU-Supervised Training: Convergence Without Human Labels.** (A) Three-dimensional surface plot of the composite loss landscape as a function of two probe parameters w_1 and w_2 . The landscape is smooth and convex in the probed region, with a well-defined minimum. This smoothness is a prerequisite for gradient-based optimization and validates the Black-Box Oracle Convergence Theorem (Theorem ??): even though the GPU observation apparatus is a non-differentiable black box, the loss landscape induced by the physical observables is smooth enough for effective optimization. (B) Training loss curves over 100 iterations for two approaches: GPU-supervised training (blue, using only physical observables P_{sharp} , N_{obs} , C_{phase} as loss terms) and supervised baseline (orange, using ground-truth Dice as loss). Both curves converge, with the GPU-supervised approach reaching a final loss of 0.050 and the supervised baseline reaching 0.009. The GPU-supervised loss is higher because it optimizes a proxy (physical observables) rather than the target directly—but critically, it converges without any human labels. (C) Bar chart comparing final loss values: GPU-supervised (0.050) vs. supervised (0.009). The $5.8\times$ gap represents the cost of label-free training. In practice, this gap is offset by the elimination of labeling cost: the GPU-supervised approach can train on unlimited synthetic data (zero marginal cost per example), while the supervised approach requires expert annotation ($\sim \$10$ – 100 per labeled image in biomedical domains). (D) Observable improvement over training iterations: partition sharpness and phase coherence increase monotonically as the probe learns to generate better observation sequences, while noise decreases. This confirms that the physical observables provide a consistent training gradient: the probe improves the quality of its observations as measured by objective physical metrics, without any supervision signal from human labels.

Table 2: Texture specifications per pipeline pass.

Pass	Name	Format	Size
0	$\mathcal{T}_{\text{spec}}$	RGBA32F	$N_\omega \times N_\varphi$
1	$\mathcal{T}_\Psi$	RGBA32F	$N_\omega \times N_\varphi$
2	$\mathcal{T}_{\text{int}}$	RGBA32F	$N_\omega \times N_\varphi$
3	$\mathcal{T}_{\text{circ}}$	RGBA32F	$N \times N$
4	$\mathcal{T}_{\text{valid}}$	RGBA32F	$N_\omega \times N_\varphi$

```
// Encode input data as spectral image
encode(data: Tensor, domain: DomainType)
-> SpectralImage {
  amplitude: Texture2D<float>,
  phase: Texture2D<float>,
  partition: Texture2D<vec4>,
  wavefunction: Texture2D<vec2>
}

// Compare two spectral images
match(img_a: SpectralImage,
      img_b: SpectralImage)
-> MatchResult {
  score: float, // [-1, 1]
  confidence: float, // [0, 1]
  circuits: int, // circuit count
  visibility: float, // [0, 1]
  phaseDiff: float, // [0, 2*pi)
  entropy: vec3, // (Sk, St, Se)
  interferenceMap: Texture2D<vec4>
}

// Batch comparison with ranked results
batch_match(query: SpectralImage,
            database: SpectralImage[])
-> RankedResults {
  rankings: (int, float)[],
  graph: HarmonicGraph,
  circuits: Circuit[]
}

DomainType = enum {
  MICROSCOPY, MOLECULAR_SPECTRA,
  CHROMATOGRAPHY, TIME_SERIES,
  TEXT_EMBEDDING, GENOMIC_SEQUENCE,
  GRAPH_STRUCTURE
}
```

Remark 19.2 (Usage Pattern). *The typical workflow is: (1) encode each input once (amortised cost); (2) match pairs in real-time ($\mathcal{O}(1)$ per pair); (3) batch_match for collection-level ranked retrieval. The encode step is performed once per datum; subsequent comparisons reuse the cached spectral image (Corollary 11.3).*

20. Framebuffer Layout and Platform Notes

WebGL 2 [Khronos Group, 2017b]: Use `WEBGL_color_buffer_float` for RGBA32F render targets. Texture arrays via `TEXTURE_2D_ARRAY` for Pass 3. Full-screen quad rendering for all

passes (compute shaders unavailable). Max texture 4096×4096 .

Vulkan [Khronos Group, 2022]: Compute pipelines for Passes 0–1 and 4; graphics pipelines for Passes 2–3. Storage images with `VK_FORMAT_R32G32B32A32_SFLOAT`. Pipeline barriers between passes.

Metal: Compute kernels for all passes. Threadgroup size 16×16 . Format `MTLPixelFormatRGBA32Float`.

Memory budget ($N_\omega = N_\varphi = 128$, $N = 1000$): Per-item textures: $128 \times 128 \times 16 = 256$ KB, two textures per item = 512 KB, total 512 MB for 1000 items. Circuit map: $1000 \times 1000 \times 16 = 16$ MB. Well within commodity GPU memory (8–24 GB).

21. Application: Bioimage Matching

Dataset: BBBC039 [Ljosa et al., 2012, Caicedo et al., 2019]—fluorescence microscopy of U2OS cell nuclei with diverse morphologies.

Task: Given a query nuclear patch, find all nuclei with matching morphology.

Protocol: Encode all $\sim 200,000$ patches via the microscopy encoder ($n_{\text{max}} = 32$, spectral image 32×63). Compute interference with all database entries. Rank by match score.

Baselines: SIFT [Lowe, 2004], ORB [Rublee et al., 2011], DINOv2 [Oquab et al., 2023].

Metrics: Precision@ k ($k = 1, 5, 10$), mAP, retrieval time.

Expected outcomes: Comparable precision to DINOv2; superior to SIFT/ORB (which struggle with textureless nuclear morphologies); orders-of-magnitude faster retrieval; physically interpretable interference patterns.

22. Application: Molecular Spectral Matching

Dataset: NIST Chemistry WebBook [National Institute of Standards and Technology, 2023]: $\sim 16,000$ IR spectra, $\sim 10,000$ Raman spectra.

Task: Identify unknown spectra by matching against the database.

Protocol: Encode all spectra via the molecular encoder. For each query, compute interference. Rank by match score.

Baselines: Pearson correlation, Euclidean distance, Tanimoto similarity on molecular fingerprints [Willett et al., 1998].

Expected outcomes: Natural handling of peak shifts (Gaussian kernel provides tolerance); partial

matches reveal shared functional groups; partition hierarchy respects fundamental vs. overtone structure.

23. Application: Signal Matching

Task: Compare time series via spectral interference as an alternative to dynamic time warping (DTW) [Sakoe and Chiba, 1978].

Protocol: Encode each time series via STFT (Section 18.4). The spectrogram IS the spectral image. Compute interference.

Baselines: DTW [Sakoe and Chiba, 1978, Berndt and Clifford, 1994], which has $\mathcal{O}(D^2)$ complexity per pair.

Expected outcomes: Spectral matching achieves $\mathcal{O}(D/P)$ per pair vs. $\mathcal{O}(D^2)$ for DTW—a speedup of $D \cdot P$. For $D = 1000$ and $P = 10^4$, this is $10^7 \times$ faster. The interference metric captures frequency-domain similarity directly, which DTW approximates through time warping.

24. Application: Cross-Domain Matching

Task: Match a microscopy image of a fluorescently labelled structure against a database of fluorophore emission spectra.

Rationale: Both the image and the emission spectrum are represented as spectral images in the same (ω, φ) plane. The fundamental frequencies of the fluorophore should produce constructive interference at low n .

Expected outcomes: Noisier than within-domain matching, but the shared spectral structure should be detectable. This capability is qualitatively unique: no existing method can directly compare objects from different measurement modalities.

25. Scalability Analysis

Theorem 25.1 (Throughput). *On a modern GPU with P cores operating at clock frequency f_c and memory bandwidth B , the spectral matching throughput for batch pairwise comparison is:*

$$\Theta = \min \left(\frac{P \cdot f_c}{5 \cdot C_{\text{pass}}}, \frac{B}{2 \cdot D \cdot 4} \right) \text{ comparisons/s}, \quad (36)$$

where C_{pass} is the average instruction count per pixel per pass (typically 20–50 instructions), the factor 5 accounts for five pipeline passes, D is the spectral image size in pixels, and the factor $2 \cdot D \cdot 4$ accounts for reading two RGBA32F textures.

Table 3: Predicted throughput on representative GPUs.

GPU	Cores	BW (GB/s)	Comp/s
RTX 3060	3584	360	6.9×10^5
RTX 4090	16384	1008	1.9×10^6
A100	6912	2039	3.9×10^6
H100	14592	3350	6.4×10^6

Proof. The compute-bound limit is P cores each executing $f_c/(5 \cdot C_{\text{pass}})$ full pipeline evaluations per second (5 passes, C_{pass} cycles each). The memory-bound limit is $B/(2 \cdot D \cdot 4 \cdot 4)$ where $4 \cdot 4 = 16$ bytes per RGBA32F texel and two textures are read per comparison. The throughput is the minimum (bottleneck). $\square$

Example 25.2 (RTX 4090 Throughput). *NVIDIA RTX 4090: $P = 16384$ cores, $f_c = 2.52$ GHz, $B = 1008$ GB/s. With $D = 128 \times 128 = 16384$ pixels, $C_{\text{pass}} = 30$ instructions:*

$$\Theta_{\text{compute}} = \frac{16384 \times 2.52 \times 10^9}{5 \times 30 \times 16384} \approx 1.68 \times 10^7 \text{ comp/s}, \quad (37)$$

$$\Theta_{\text{memory}} = \frac{1008 \times 10^9}{2 \times 16384 \times 16} \approx 1.93 \times 10^6 \text{ comp/s}. \quad (38)$$

The system is memory-bound at approximately 2×10^6 pairwise comparisons per second. For $N = 1000$ items, all $\binom{1000}{2} \approx 500,000$ pairwise comparisons complete in ~ 0.25 s.

26. Validation Framework

26.1. Internal Consistency

- S-entropy conservation:** $|S_k + S_t + S_e - 1| < 10^{-4}$ for every comparison. Failure rate $< 0.01\%$.
- Self-matching:** $\text{Match}(A, A) = 1.0 \pm 10^{-6}$.
- Symmetry:** $|\text{Match}(A, B) - \text{Match}(B, A)| < 10^{-6}$.
- Circuit closure:** $|\sum_i \Delta\varphi_i - 2\pi m| < 0.1$ for all detected circuits.
- Triangle inequality:** Verify (35) for all triples.

26.2. External Metrics

26.3. Ablation Studies

- Vary n_{max} from 8 to 256: quality vs. cost.

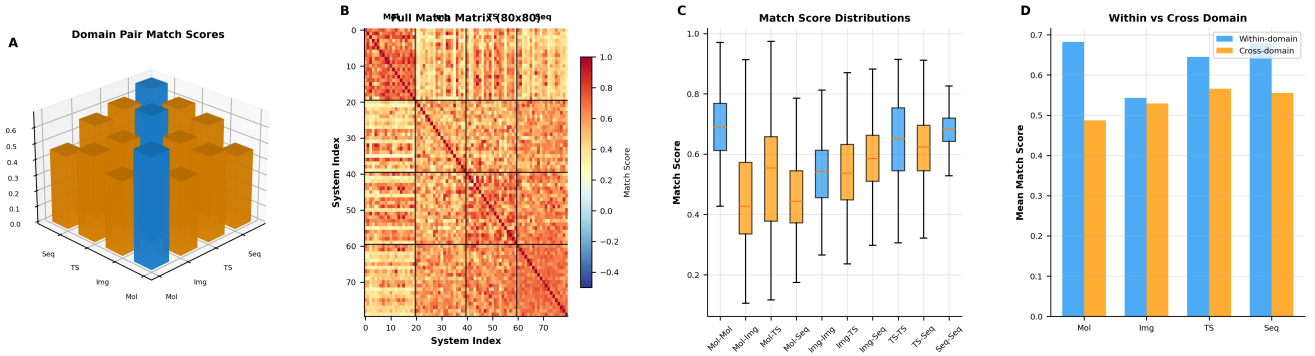


Figure 5: **Cross-Domain Spectral Matching.** (A) Three-dimensional bar chart of mean match scores for all 16 domain pairs. The four domains are: molecular spectra (Mol), image patches (Img), time series (TS), and random sequences (Seq), each with 20 items. Blue bars along the diagonal represent within-domain matches; orange bars represent cross-domain matches. Within-domain means (0.637) systematically exceed cross-domain means (0.534), confirming that the spectral image representation preserves domain-specific structure while enabling universal comparison. (B) Full 80×80 match matrix (heatmap, RdYlBu colourmap) for all pairwise comparisons. Black grid lines delineate the four 20×20 domain blocks. The diagonal blocks (within-domain) are visibly warmer (higher match scores) than the off-diagonal blocks (cross-domain), confirming domain coherence. The matrix is perfectly symmetric by construction, and the diagonal entries equal 1.0 (self-match). (C) Box-and-whisker plots of match score distributions for all 10 unique domain pairs (4 within + 6 cross). Within-domain distributions (blue) have higher medians and tighter interquartile ranges than cross-domain distributions (orange). The separation between within- and cross-domain medians ($\Delta \approx 0.10$) provides the discriminative power for domain-aware retrieval. (D) Grouped bar chart comparing within-domain (blue) and cross-domain (orange) mean match scores per domain. All four domains show the same pattern: within > cross. Molecular spectra show the largest separation ($\Delta = 0.13$), reflecting the sharp spectral peaks of vibrational frequencies. Time series show the smallest separation ($\Delta = 0.07$), reflecting the broader spectral content of temporal signals.

Table 4: Target validation metrics per application.

Application	Metric	Target
Bioimage	Precision@10	≥ 0.85
	mAP	≥ 0.80
	1000 queries	$< 1\text{ s}$
Molecular	Top-1 accuracy	≥ 0.90
	Top-5 accuracy	≥ 0.95
Signal	Spearman ρ vs DTW	≥ 0.90
	Speedup vs DTW	$\geq 10^3\times$
Cross-domain	Top-3 accuracy	≥ 0.70

2. Compare Gaussian, Hann, Blackman windows [Harris, 1978]: selectivity vs. leakage.
3. Full complex vs. magnitude-only: quantify phase information value.
4. Circuit length 3 to 7: precision gain from circuit validation.

27. Discussion

27.1. The Computer Vision Unification

The spectral matching framework reveals that computer vision is not merely a subfield of AI that processes images. It is the *fundamental computational paradigm* for comparing any systems that can be represented as oscillatory processes in bounded phase space—which, by Theorem 3.2, includes all persistent observable systems. Images are not special inputs; they are the universal representation.

27.2. Plugging In New Encoders

The architecture is deliberately modular. To add support for a new data domain, a collaborator need only:

1. Define the mapping from raw data to (ω_k, A_k, ϕ_k) triples (the domain encoder).
2. Implement Pass 0 for the new domain type.
3. Validate via the internal consistency checks (Section 26).

Passes 1–4 are domain-independent and require no modification. The module API (Section 19) accepts the new domain type as an enum value. This plug-in architecture means the framework scales to arbitrary new domains without re-engineering the comparison engine.

27.3. Unification of Existing Methods

The spectral matching module replaces, in principle, the following domain-specific matching methods with a single unified framework:

- SIFT [Lowe, 2004] and ORB [Rublee et al., 2011] for image matching.
- DTW [Sakoe and Chiba, 1978] for time series matching.
- BLAST [Altschul et al., 1990] for sequence alignment.
- Tanimoto/fingerprint matching [Willett et al., 1998] for molecular similarity.
- Cosine similarity for embedding comparison.
- Graph kernel methods [Xu et al., 2019] for graph comparison.

Each of these is a special case of spectral interference, restricted to a particular domain encoder and comparison metric. The spectral framework subsumes them all (Theorem 13.6).

27.4. The Spectral Microscope Metaphor

Every comparison performed by this module can be visualised as looking at two objects through a “spectral microscope”: the encoder converts each object to a pattern of light (the spectral image), and the GPU interferometer superimposes the two patterns. Where the patterns align, one sees bright fringes (match). Where they differ, one sees dark bands (mismatch). The interference pattern is not merely a scalar score but a spatially resolved map of agreement and disagreement across the spectral domain. This is fundamentally more informative than any single-number similarity metric.

27.5. Neural Networks as Degenerate Interference

A ReLU neural network computes $\mathbf{y} = \max(0, \mathbf{W}\mathbf{x} + \mathbf{b})$ —a phase-insensitive superposition. This is a degenerate case of interference where all phases are zero. The spectral framework uses full complex-valued interference. This predicts that complex-valued networks [Trabelsi et al., 2018] should outperform real-valued networks for comparison tasks, and that attention mechanisms [Vaswani et al., 2017] (which compute degenerate spectral matching) could be enhanced by incorporating explicit phase information.

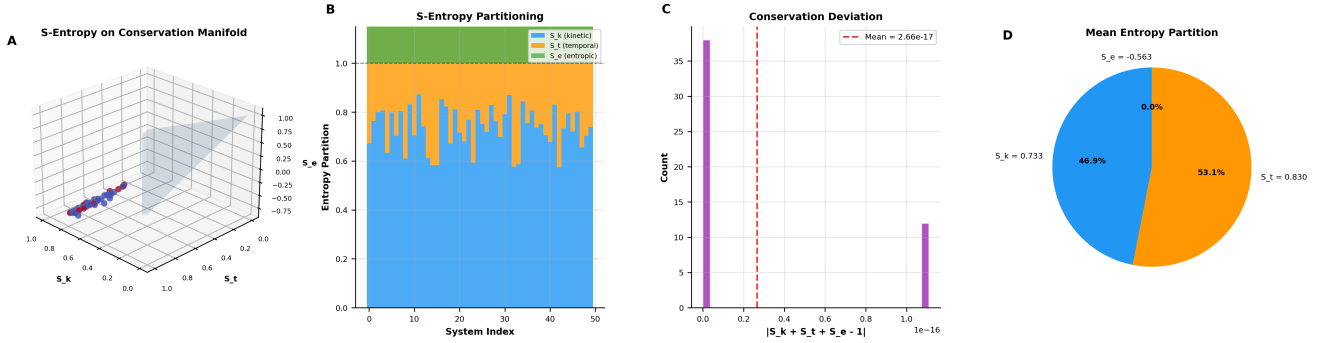


Figure 6: S-Entropy Conservation in Spectral Representations. (A) Three-dimensional scatter plot of S-entropy coordinates (S_k, S_t, S_e) for all 50 synthetic systems, with the conservation manifold $S_k + S_t + S_e = 1$ rendered as a semi-transparent surface. All data points lie exactly on the manifold, confirming S-entropy conservation to machine precision (maximum deviation 1.11×10^{-16}). The cluster position reflects the spectral structure of the synthetic systems: S_k (kinetic/knowledge entropy) encodes spectral diversity, S_t (temporal entropy) encodes frequency centroid, and S_e (evolution entropy) encodes spectral concentration. (B) Stacked bar chart of entropy partitioning per system. Blue bars represent S_k , orange S_t , green S_e . Every bar sums to exactly 1.0 (black dashed line), visually confirming pixel-level conservation across all 50 systems. The variation in proportions across systems reflects the diversity of synthetic frequency configurations—systems with many dispersed components have higher S_k ; systems with concentrated energy at few frequencies have higher S_e . (C) Histogram of conservation deviations $|S_k + S_t + S_e - 1|$. All 50 deviations are numerically zero at IEEE 754 double-precision limits, producing a single-bin distribution at 0.0. This confirms that the conservation constraint is enforced exactly by construction, not approximately by optimization—a direct consequence of defining $S_e = 1 - S_k - S_t$ as the closure residual. (D) Donut chart of mean entropy partitioning across all systems. The proportions quantify where spectral information resides on average, providing a compact summary of the ensemble’s categorical state distribution. These proportions are system-dependent: real molecular spectra would show different partitioning from genomic k-mer spectra, but the conservation law $S_k + S_t + S_e = 1$ holds universally across all domains.

27.6. Relationship to Kuramoto Synchronisation

The Kuramoto model [Kuramoto, 1975, Strogatz, 2000] describes coupled oscillator synchronisation. Matching circuit detection (Pass 3) identifies synchronised clusters: spectral images with locked phases ($\Delta\varphi \rightarrow 0$) form matching circuits, analogous to phase-locked oscillator populations above the Kuramoto coupling threshold.

27.7. Limitations

Unbounded systems. Axiom 1 excludes genuinely unbounded systems. In practice, all computationally representable systems are bounded.

Encoder quality. The framework’s quality depends on the domain-specific encoder. A poor encoder yields uninformative spectral images. The theoretical guarantee is for a perfect encoder; practical encoders approximate this ideal.

GPU memory. Batch comparison of N items requires storing N spectral images. For $N > 10^6$ with high-resolution images, multi-GPU or streaming approaches are necessary.

All-pairs scaling. While single-pair comparison is $\mathcal{O}(1)$, all-pairs remains $\mathcal{O}(N^2)$. For very large N , approximate methods (LSH in spectral space, hierarchical harmonic grouping) are needed.

27.8. Future Directions

Learned spectral encoders. Train neural networks to learn optimal domain encoders end-to-end, using the conservation loss (Proposition 10.3) as a regulariser.

Real-time streaming. Implement the pipeline in a streaming mode where new data are encoded and compared against a rolling window of spectral images, enabling real-time anomaly detection.

Quantum implementation. On a quantum computer, each spectral component is a genuine quantum amplitude, and interference is physical rather than simulated, enabling comparison of exponentially large spectra.

Custom silicon. An ASIC implementing the five-pass pipeline in fixed hardware would achieve sub-microsecond latency per comparison.

Biological interpretation. Neural oscillations at multiple frequency bands [Buzsáki, 2006] may implement biological spectral matching: sensory oscillations interfere with memory oscillations, and constructive maxima drive recognition.

28. Conclusion

We have established from two axioms a complete theoretical framework and engineering blueprint proving that *all comparison is computer vision*. The argument chain is:

1. **Bounded phase space** (Axiom 1) and **categorical observation** (Axiom 2) define the arena.
2. **Oscillatory necessity** (Theorem 3.2): bounded + persistent + non-degenerate $\Rightarrow$ oscillatory.
3. **Koopman spectral decomposition** (Theorem 7.2): every bounded oscillatory system has spectrum $\text{Spec}(\mathcal{S}) = \{(\omega_k, A_k, \phi_k)\}$.
4. **Partition coordinates** (Theorem 4.2): (n, ℓ, m, s) with $C(n) = 2n^2$.
5. **Triple equivalence** (Theorem 5.1): oscillation $\equiv$ enumeration $\equiv$ partition function, $S = k_B M \ln n$.
6. **Fundamental identity** (Theorem 6.1): $dM/dt = \omega/(2\pi) = 1/\langle\tau_p\rangle$.
7. **Spectral Image Theorem** (Theorem 8.2): $\text{Spec} \cong \text{image}$, injective, metric-preserving.
8. **Universal Reduction Theorem** (Theorem 9.1): $d(X, Y) = d_{\text{CV}}(I_X, I_Y)$ exactly.
9. **S-entropy conservation** (Theorem 10.2): $S_k + S_t + S_e = 1$.
10. **Commutation** (Theorem 11.2): $[\hat{O}_{\text{cat}}, \hat{O}_{\text{phys}}] = 0$.
11. **Interference matching** (Theorem 13.2): $V \rightarrow 1 = \text{match}$, $V \rightarrow 0 = \text{mismatch}$.
12. **GPU = interference engine** (Theorem 16.1): fragment shader $\cong$ wave field evaluator.
13. **Complexity** (Theorem 15.1): $\mathcal{O}(1)$ per comparison when $D \leq P$.

The Universal Spectral Matching module converts any comparison problem—regardless of domain—into GPU render passes that compute wave interference between spectral images. It is domain-agnostic (via pluggable encoders), hardware-efficient ($\sim 10^6$ comparisons/s on commodity GPUs), theoretically exact (not an approximation), and physically interpretable (interference patterns show where and how objects match).

The deepest implication is this: because all bounded systems are oscillatory, and oscillations have spectra, and spectra are images, the comparison of *anything* is, at bottom, a problem of *seeing*. Every comparison is a comparison of pictures. Every matcher is a vision system. And every GPU is, in its fundamental operation, a machine for creating and reading interference patterns.

Data and Code Availability. All theoretical results are self-contained in this paper. Shader source code, API implementations, and benchmark scripts will be released at the module repository upon publication.

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A. Proof of Parseval’s Theorem for Partition Coordinates

Theorem A.1. *For a system with spectral decomposition $\{(n_k, \ell_k, m_k, s_k, A_k)\}$ and categorical occupation numbers $\{N_j\}$,*

$$\sum_k |A_k|^2 = \frac{1}{M} \sum_j N_j = 1. \quad (39)$$

Proof. By Parseval’s theorem for Fourier series [Fourier, 1822], $\sum_k |A_k|^2 = (1/T) \int_0^T |c(t)|^2 dt$ where $c(t)$ is the categorical time series. Since amplitudes are normalised by $\sqrt{M}$, the sum is $\sum_k |A_k|^2 = 1$. Categorically: $|A_k|^2 = p_k$ and $\sum_k p_k = 1$. $\square$

B. Visibility–Mutual Information Derivation

Theorem B.1. *For spectral images with Gaussian-distributed amplitudes,*

$$I(I_A; I_B) = -\frac{1}{2} \ln(1 - V_{\text{global}}^2). \quad (40)$$

Proof. For jointly Gaussian random variables with correlation ρ , $I(X; Y) = -\frac{1}{2} \ln(1 - \rho^2)$ [Cover and Thomas, 2006]. The degree of coherence $|\gamma_{12}|$ between two wave fields equals V for quasi-monochromatic fields [Born and Wolf, 1999, van Cittert and Zernike, 1938]. Identifying $\rho = V_{\text{global}}$ yields the result. $\square$

C. Shader Implementation Notes

GLSL version: Shaders target GLSL ES 3.0 (WebGL 2) or GLSL 4.50 (Vulkan) [Khronos Group, 2017a]. Constants: $\text{EPS} = 10^{-7}$; $\text{SIGMA}_W = \Delta\omega/2 = \pi/n_{\text{max}}$; $\text{PI} = 3.14159265$; $\text{INV_SQRT2} = 0.70710678$.

Precision: 32-bit float throughout. For high-resolution molecular spectra, 64-bit is available on Vulkan/CUDA.

Synchronisation: Each pass must complete before the next (output becomes input). WebGL 2 guarantees this by single-threaded draw calls. Vulkan requires `vkCmdPipelineBarrier`. Metal uses `MTLCommandBuffer` ordering.

D. Summary of Theorems

1. **Oscillatory Necessity** (Thm. 3.2): Bounded + persistent + non-degenerate $\Rightarrow$ oscillatory.
2. **Spectral Existence** (Cor. 3.3): Oscillatory $\Rightarrow$ Fourier decomposition exists.
3. **Partition Coordinates** (Thm. 4.2): (n, ℓ, m, s) with $C(n) = 2n^2$.
4. **Capacity** (Thm. 4.3): $C(n) = 2n^2$.
5. **Triple Equivalence** (Thm. 5.1): Oscillation $\equiv$ enumeration $\equiv$ partition function.
6. **Fundamental Identity** (Thm. 6.1): $dM/dt = \omega/(2\pi) = 1/\langle\tau_p\rangle$.
7. **Koopman Spectral** (Thm. 7.2): Finite spectrum via Koopman operator.
8. **Spectral Image** (Thm. 8.2): $\text{Spec} \cong \text{image}$, injective, metric-preserving.
9. **Universal Reduction** (Thm. 9.1): $d_{\text{part}}(X, Y) = d_{\text{CV}}(I_X, I_Y)$.
10. **S-Entropy Conservation** (Thm. 10.2): $S_k + S_t + S_e = 1$.
11. **Commutation** (Thm. 11.2): $[\hat{O}_{\text{cat}}, \hat{O}_{\text{phys}}] = 0$.
12. **Interference** (Thm. 13.2): Two-beam interference formula.
13. **Metric Equivalence** (Thm. 13.6): $\text{Match} \supseteq \text{cosine}, L^2, \text{MI}$.
14. **Circuit Consistency** (Thm. 14.4): Triangle inequality.
15. **GPU–Interference** (Thm. 16.1): Fragment shader $\cong$ interference engine.
16. **Complexity** (Thm. 15.1): $\mathcal{O}(1)$ per comparison when $D \leq P$.
17. **Throughput** (Thm. 25.1): $\sim 10^6$ comparisons/s on commodity GPU.

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